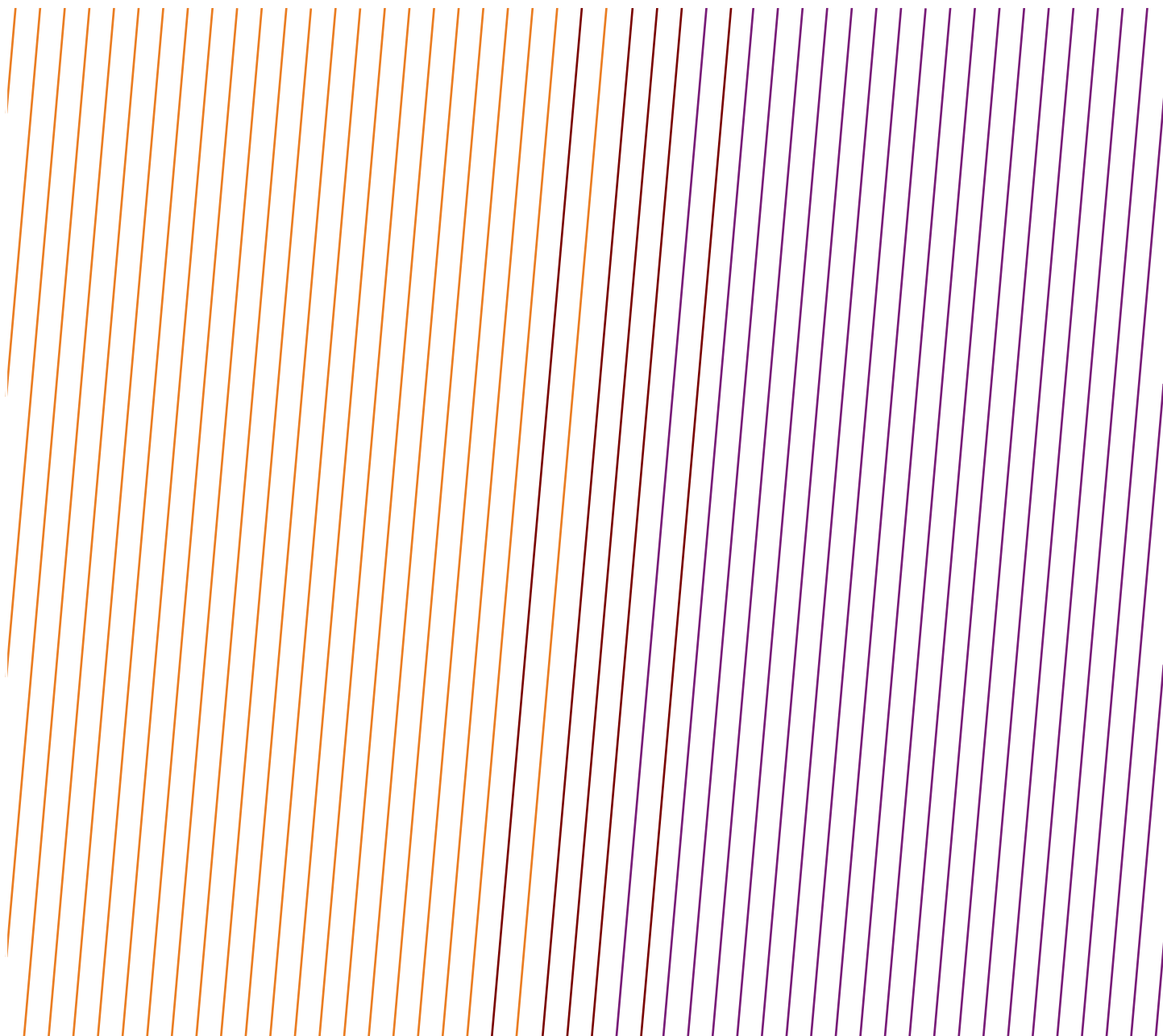


DATA SERIES

Safety performance indicators – 2022 data – High potential event reports



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This report was produced by the Safety Committee.

Feedback

IOGP welcomes feedback on our reports: publications@iogp.org

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DATA SERIES

Safety performance indicators – 2022 data – High potential event reports

Revision history

VERSION	DATE	AMENDMENTS
1.0	June 2023	First release

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AFRICA ONSHORE

DATE: 06 Apr 2022

COUNTRY: Algeria

FUNCTION: Drilling

CAUSE: Caught in, under or between (excl. dropped objects)

ACTIVITY: Lifting, crane, rigging, deck operations

RULE: Safe mechanical lifting

NARRATIVE:

A forklift overturned while assisting the drilling contractor with extending the mast raising line.

WHAT WENT WRONG:

The operation was a rig move (dismantling the rig parts and send them to the new location). The third-party forklift operator was assisting the drilling contractor with extending the mast raising lines to be inspected by third party company inspectors in the old location before moving to the new location. The first raising line extended safely in the proper place ready for the inspection. While continuing with the second raising line, suddenly, the forklift overturned on its right side and sustained damage.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

The assigned job supervisor (assistant driller) left and kept only the floorman with the forklift operator where he failed to use stop work authority, and failed also to assess the operation risks.

CAUSAL FACTORS:

PEOPLE (ACTS): Use of Tools, Equipment, Materials and Products: Improper use/position of tools/equipment/materials/products

PEOPLE (ACTS): Inattention/Lack of Awareness: Lack of attention/distracted by other concerns/stress

PROCESS (CONDITIONS): Organizational: Inadequate supervision

DATE: 22 Jun 2022

COUNTRY: Algeria

FUNCTION: Drilling

CAUSE: Dropped objects

ACTIVITY: Lifting, crane, rigging, deck operations

RULE: Safe mechanical lifting

NARRATIVE:

Near miss: Slickline tool dropped.

During second run of slickline operations and after setting the lock mandrel in the landing nipple, the contractor operator was in a slickline unit pulling out a toolstring using a joystick handle.

When the toolstring reached the surface, the wire did not stop and the toolstring continued going up at full velocity. Therefore, the toolstring hit the upper sheave wheel and the rope socket (connection between wire and toolstring) parted, causing the toolstring to drop from approximately 25 m height and onto the rig floor.

No one was on the rig floor when, the incident happened.

AFRICA ONSHORE

WHAT WENT WRONG:

- The supervisor from the third-party company stepped out for a break and the supervisor assistant was operating the slickline unit.
- The supervisor did not set alarms in the electronic depth control system.
- The supervisor was not trained on this AMS system.
- Overconfident.
- Human error.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

- Reinforce importance of “focus” and “attention” by personnel operating machinery during pre-job safety meetings.
- Pre-job briefing meetings between slickline field crew and technical support engineer/field service manager must cover the setting of control measures to prevent uncontrolled toolstring release from wellbore.
- Slickline field personnel on supervisory role must meet minimum competence criteria. Slickline service provider must guarantee such competence when deploying personnel to the field.
- Slickline service provider to audit that slickline field crews adhere to operational procedures by reviewing job logs retrieved from the unit.
- Pre-set alarms are mandatory prior to start any job to prevent human errors due to over confidence.
- Effective Management of red zone area can prevent serious injuries.

CAUSAL FACTORS:

No Causal Factors Allocated

DATE: 23 Oct 2022

COUNTRY: Algeria

FUNCTION: Production

CAUSE: Pressure release

ACTIVITY: Production operations

RULE: Bypassing safety controls

NARRATIVE:

There was an Emergency Total Shutdown of the central processing facility. The root cause of this trip was not known, and so electrical and instrument supervisors were brought in to check if there was any problem before proceeding to restart the facilities. During this period and exactly at 06:45 there was gas detection from several gas detectors. A cloud of gas was noticed coming from the old horizontal flare pit. All production wells were immediately shutdown and the facilities depressurized.

HSE fire brigade installed water curtains and opened several fire monitors to dilute and disperse the gas cloud. After closing the production wells and depressurizing the facilities, the gas cloud started decreasing until it had disappeared. No Personal injury or asset damage resulted from this event.

WHAT WENT WRONG:

Backflow from hydrocarbon source. Check valves were the only protection layer and they failed due to internal presence of solids.

Water dilution line connected to the decommissioned horizontal flare pit, due to a previous non-conformant MoC.

Process Safety Fundamental: We maintain safe isolation.

AFRICA ONSHORE

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

Lack of as built drawings.

CAUSAL FACTORS:

No Causal Factors Allocated

DATE: 22 May 2022

COUNTRY: Cameroon

FUNCTION: Production

CAUSE: Exposure electrical

ACTIVITY: Excavation, trenching, ground disturbance

RULE: Bypassing safety controls

NARRATIVE:

While working on digging the foundation for the perimeter fence, one of the technicians touched an underground cable with his shovel spade and saw a spark..

WHAT WENT WRONG:

Missing Documentation:

As per the standard work procedure the underground services network drawing was to be consulted to locate any buried element.

However, such a document does not currently exist. Moreover, the electricity company stipulated not to disclose their cables layout to public.

Inadequate Hazard Control:

Warning signs (plastics nets; white sand; signage boards etc.) of underground services and site drawing indicating underground services were absent leaving staff with vigilance as the only control measure to continue trenching activities.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

- A full mapping of underground services networks, including electrical, telecommunication, security, water and gas lines was established.
- Relocating the power lines out of area to ascertain safe continuation of project and subsequent projects and update of drawings accordingly.
- A procedure for excavation works put in place.
- Scanning devices for underground services acquired and staff trained on proper usage.

CAUSAL FACTORS:

PEOPLE (ACTS): Use of Tools, Equipment, Materials and Products: Servicing of energized equipment/inadequate energy isolation

PROCESS (CONDITIONS): Organizational: Inadequate work standards/procedures

PROCESS (CONDITIONS): Organizational: Inadequate hazard identification or risk assessment

AFRICA ONSHORE

DATE: 20 Feb 2022

COUNTRY: Congo

FUNCTION: Production

CAUSE: Explosion, fire or burns

ACTIVITY: Maintenance, inspection, testing

RULE: Bypassing safety controls

NARRATIVE:

During a rainy period, a closed drainage valve caused rainwater to accumulate on the roof of a tank that favoured the growth of moss and lichen.

There were also traces of paint on the tank roof following recent waterproofing work.

A team of operators (T1) had been working on the roof of the tank to clean off the moss and traces of paint. The team was using a presumed algaecide/fungicide considered as a specific, multi-surface, grease-removing agent – and equipment (PPE, scrapers, brooms, etc.).

Later, a second team (2), which was working on the tank wall, noticed a paint run in their work area. Because they did not have any cleaning agent, they asked team (1) to give them some of the product. Team (2) took what they thought was an empty metal pot of hardener as a recipient for the cleaning agent, in fact containing a residue of hardener and paint from a previous work – and poured part of the cleaning agent into it. When they had finished their task (around midday), the rest of the product that had not been used (around 3 litres) was handed back to team (1).

At the end of the day, the remaining product in the metal pot was poured back into a container of the pure product. Exhausted by the heat, the operators left the work area, leaving the equipment and products on site.

In the early evening, two firefighters performing their round noticed smoke coming from the tank roof and stepped in to contain the fire outbreak. The container into which the product had been poured had melted and spontaneously combusted. The fire had then spread to the equipment nearby and damaged part of the separating wall on the periphery of the roof. Immediate actions: -Intervention by firefighters and triggering of the deluge system of the tank.

WHAT WENT WRONG:

- Exothermic reaction after chemicals had been mixed (paint +product + hardener) in a closed polyethylene container.

Additional causes identified:

- Poor chemical storage management.
- Work area not properly tidied at the end of the shift.
- It was Insufficient maintenance of the rainwater drainage system causing moss to accumulate.
- Insufficient supervision of works and non-compliance with work permit close-out rules.
- Lack of knowledge and awareness of the risks related to the use of chemicals.
- Fatigue due to heat exposure. Conditions not identified during the risk assessment.

Process Safety Fundamental: We control ignition sources. We respect hazards.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

- Check that the operators handling chemicals have received suitable training.
- During safety rounds, check that chemical containers are suitable, properly labelled and stored in areas surrounded by a means of retention.

AFRICA ONSHORE

- Factor in information on chemical compatibility (Ref. Material Safety Data Sheets).
- Remind people of the obligation to clean and tidy the work area when work is finished, preferably when the permit is closed out for the day.
- Check that the tank rainwater drainage valves are open during rainy periods and/or after periods of heavy rainfall.
- Remind team managers about the work permit process.
- Take into account the stress involved in the work and the exposure to intense heat when preparing the work and in the risk assessment.

CAUSAL FACTORS:

PEOPLE (ACTS): Use of Tools, Equipment, Materials and Products: Improper use/position of tools/equipment/materials/products

PEOPLE (ACTS): Inattention/Lack of Awareness: Fatigue

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate design/specification/management of change

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate maintenance/inspection/testing

PROCESS (CONDITIONS): Organizational: Inadequate training/competence

DATE: 13 Jan 2022

COUNTRY: Gabon

FUNCTION: Drilling

CAUSE: Dropped objects

ACTIVITY: Lifting, crane, rigging, deck operations

RULE: Safe mechanical lifting

NARRATIVE:

At 15h42, an incident occurred on the Well head (WH) during a routine scraping operation. During the rig down phase, the truck hoisting winch cable broke when lifting the Blow Out Preventer (BOP) from the well head (WH) until its receptacle on the mast. The pressure control equipment (PCE), no longer being held by the lifting cable struck the gangway surrounding the WH and the lubricator tipped over the gangway access ramp to the ground. The gangway guardrails were bent and part of the PCE damaged. The operator standing on the gangway at the BOP side was slightly injured on his right leg by the guide rams. At the clinic, he received some first aid treatment for few minutes and returned to his job.

A truck was requested for removing the PCE from the platform, then the well was secured and put back into production.

WHAT WENT WRONG:

- Truck's design (without safety restrain slings, anti -2-block system and Crane Override System) creates control's constraints.
- Size of the wellhead not suitable for the full application of the procedure.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

- Implement Preventive Maintenance & Inspection of winch cables with regular intervals.
- Follow the Line of Fire safety rules. Keep yourself and others out of the Line of Fire.

AFRICA ONSHORE

- Follow the safe mechanical lifting safety rules. Plan lifting operations and control the area.
- Always comply with procedures. Noncompliance is subject to Consequence Management.
- Update the scraping procedure in accordance with Well Heads platforms design. All operational and situational aspects of well heads including platforms are to be taken into account and captured accordingly.

CAUSAL FACTORS:

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate maintenance/inspection/testing

DATE: 14 Sep 2022

COUNTRY: Gabon

FUNCTION: Production

CAUSE: Explosion, fire or burns

ACTIVITY: Production operations

RULE: Other

NARRATIVE:

A wax re-injection project was commissioned to solve the problem of the accumulation of more than 2,000 bbl. of wax at the landfill for several years, by purchasing and installing an electric heater in a merlon to dissolve the wax and re-inject it into the process. On September 14, around 8:15 a.m., a contractor noticed a fire starting inside the wax dissolving merlon located at the level of the pig receiver of the 18" pipeline outlet. The contractor hit the nearby fire extinguisher (water and additive) and then went to the control room to raise the alarm.

WHAT WENT WRONG:

- The Procedure was adapted to the project, but has shortcomings for the management of wax from the 18" because the level estimates are done by eye.
- The personnel had not been sufficiently trained on the new wax heating system.

Process Safety Fundamental: We control ignition sources. We respect hazards.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

- Revise the wax reinjection procedure to emphasize the fire hazard and mitigation measures.
- Install a low-level safety device in the merlon.
- Implement a weekly training program by the fire team.
- Re-evaluate the use of fire extinguishers in our facilities.
- Investigate the possibility of an alternative means of processing wax.
- Strengthen the procedure for awarding Projects to Operations
- Indicate which HAZID/HAZOP/MoC. procedures are required depending on the type of project.
- Re-evaluate the ordering process for Safety Critical equipment and tools by involving the SSE department.

CAUSAL FACTORS:

PROCESS (CONDITIONS): Protective Systems: Inadequate security provisions or systems

AFRICA ONSHORE

DATE: 06 Nov 2022

COUNTRY: Gabon

FUNCTION: Production

CAUSE: Explosion, fire or burns

ACTIVITY: Production operations

RULE: Unspecified

NARRATIVE:

At approximately 1:25 a.m., an emergency shutdown of the facility occurred. At the same time, an operator in the area on tour noticed a fire at the crude export pumps. The operator immediately alerted the control room, which triggered the emergency response (1111).

The night shift (three staff members) went to the muster point 50 m from the station, as required by the emergency procedure. At approximately 1:35 a.m., the First Response Team (FIT) arrived at the station and after a quick assessment of the risks and the extent of the flames, they decided to attack the fire safely, using the fire extinguishers present in the station. It should be noted that 100% of the fire extinguishers at the site were used and replaced within three days.

WHAT WENT WRONG:

Operational Risk Assessment (ORA) procedure is available, but not understood and even less rigorously applied on site.

Poor management of operational changes on site without technical authorities.

Process Safety Fundamental: We respect hazards.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

- Organize awareness and training sessions on operational risk management procedures. Emphasizing the risks associated with all operational changes.
- Follow recommendations from previous similar incidents.
- Strengthen the risk assessment process for any change or validation of a project or technical solution – All operational changes must be validated by the discipline's technical authority.
- Re-evaluate the ordering process for safety critical equipment and spare parts by involving the technical authorities.
- A Company expectation is to act responsibly and safely: This is what our operations team demonstrated by attacking the fire in a stage where they felt safe to act.
- To operate safely is a Company expectation. Safety is a priority and cannot be compromised.

CAUSAL FACTORS:

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate maintenance/inspection/testing

AFRICA ONSHORE

DATE: 16 Sep 2022

COUNTRY: Gabon

FUNCTION: Production

CAUSE: Pressure release

ACTIVITY: Production operations

RULE: Other

NARRATIVE:

In the early morning of Sept 16, 2022, at the processing facility at approximately 00:43 am, a loud noise was heard by the night panel operator, quickly followed by a total shutdown of the station (ESD1). An inspection of the processing facility revealed a damaged piece of equipment was blown 12 m further from the gas compressor. A damage assessment of other equipment was realized, and compressor isolated from the process.

WHAT WENT WRONG:

- Bolts broke by fatigue due to exceeding the running time for these parts.
- No proper monitoring of equipment running hours.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

- Continuous improvement is required for the safety of our operations. The supplier of the compressor has been involved in a review of the design Modification for the damaged equipment to improve the reliability.
- In the Company Values, we must operate responsibly and safely. This incident reminds us to pay attention while working with heavy machinery. New daily visual inspection has been put in place to detect these defects as soon as possible. A maintenance preventive plan has been updated accordingly.
- We must be flexible. A reflection has started between process safety team and site operations to rethink our compressor area in order to have a safer working environment.

CAUSAL FACTORS:

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate maintenance/inspection/testing

AFRICA OFFSHORE

DATE: 28 Jun 2022

COUNTRY: Angola

FUNCTION: Production

CAUSE: Pressure release

ACTIVITY: Maintenance, inspection, testing

RULE: Line of fire

NARRATIVE:

2 1/2" Valve separated from a fire hydrant during deluge testing and landed 9m away.

WHAT WENT WRONG:

Fire hydrant design utilized threaded connection for piping size (2.5") above recommended maximum (1.5").

Valve upstream found partially open instead of in normally closed position.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

Threaded 2.5" connections not recommended in this service - recommend back-welding flange connection or provide secondary retention until welding complete.

Survey upstream valves to ensure in normally closed position.

CAUSAL FACTORS:

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate design/specification/management of change

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate maintenance/inspection/testing

DATE: 14 Aug 2022

COUNTRY: Angola

FUNCTION: Production

CAUSE: Water related, drowning

ACTIVITY: Transport - Water, incl. marine activity

RULE: Bypassing safety controls

NARRATIVE:

During a transfer operation of personnel to the platform boat landing by boat, a wave suddenly appeared and unbalanced the boat. As a result, a contractor employee lost his balance when he was boarding on the stairs to the platform, having caught his right foot between the last step of the boat landing and the boat, causing an injury to the front of the right foot.

Immediate action: The employee was assisted onsite by colleagues and transferred for first aid treatment provided by the doctor. Due to the suspicion of a fracture, he was referred to hospital for further assessment.

Following investigation, the cause was found to be a failure to comply with the embarkation and disembarkation procedure at the boat landing. A video showed that the employee took too long to go up the stairs. The boat assistance in position of letting the waves coming behind. The Captain should have used the 2 option boat landing to control the very smooth waves of that day. Recommendations were set up and implemented to avoid reoccurrence.

AFRICA OFFSHORE

WHAT WENT WRONG:

Main root Cause: Failure to comply with the embarkation procedure at the boat landing.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

The video showed that the employee took so long to climb the stairs, boat assistance from the boat turned her back in the position of the waves, Captain should have used the 2-port option to control the very gentle waves that day.

1. The departure from boat landing to the boat and vice versa must only be done when the wave is on top.
2. The passenger must follow the guidance of the boat's assistant, sailor.
3. Video awareness session must be made to passengers during navigation.
4. Passengers must be quick when getting on and off the boat landing.
5. Produce flyers with wording such as "be careful when going up and down the boat landing your life can cost with the risk of injury or death", and install on all boat landings.
6. The sailor must position himself to see the movement of the waves to better assist passengers to get off the boat for boat landing.
7. The boat pilot must guide and implement stop work authority measures.
8. Provide adequate training for boat attendants and all passengers.

CAUSAL FACTORS:

PEOPLE (ACTS): Use of Protective Methods: Failure to warn of hazard

PEOPLE (ACTS): Inattention/Lack of Awareness: Improper decision making or lack of judgment

PEOPLE (ACTS): Inattention/Lack of Awareness: Lack of attention/distracted by other concerns/stress

DATE: 16 Feb 2022

COUNTRY: Cameroon

FUNCTION: Production

CAUSE: Struck by (not dropped object)

ACTIVITY: Lifting, crane, rigging, deck operations

RULE: Safe mechanical lifting

NARRATIVE:

A crane operator was running the crane to support the construction project team in the installation of the Mobile crane at a Platform. A signal was given to the crane operator to lower the load block and pick up the king post brace.

The load (king post brace) was successfully picked and positioned to the MAST Top section of the king post. In the attempt to socket the king post brace into the mast swivel collar, the team noticed that the upper male clevis of the King post brace would not fit into the swivel collar.

The banksman gave a signal to the crane operator to swing right, he successfully swung right and cleared the brace from the MAST top section.

The next signal was given to the crane operator to lift the load (King post brace). While the load was hoisting, the crane boom suddenly swung to the left and the load moved above the MAST Top section of the King post.

Two employees were standing on the scaffold around the MAST Top section of the King post.

AFRICA OFFSHORE

WHAT WENT WRONG:

Insufficient equipment familiarization:

- The Sr Crane Operator was used for the mobile crane installation job.
- The Sr. Crane Operator had given little or no hands-on experience to the newly hired Crane Operator.
- While the new Crane Operator was hoisting the load with the main line, the lever came out from the socket.
- He panicked and inadvertently swung the crane boom to the left and the load abruptly moved above the MAST top section of the King Post, where two employees were positioned on the scaffold.
- The lever has a stop pin to prevent it from being inadvertently unsocketed, and the pin could be tightened.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

- Establishment of a fit-for-purpose hands-on familiarization of offshore cranes for all new crane operators.
- Signage installed in the crane cabin to remind the operator of the use of the stop pin, and reiterate the importance of fastening the lever to the linkage.
- Field crane operator portfolio needs to be upgraded to include more people.

CAUSAL FACTORS:

PEOPLE (ACTS): Following Procedures: Deviation unintentional (by individual or group)

PROCESS (CONDITIONS): Organizational: Inadequate training/competence

PROCESS (CONDITIONS): Organizational: Inadequate supervision

ASIA/AUSTRALASIA ONSHORE

DATE: 24 Jan 2022

COUNTRY: Indonesia

FUNCTION: Construction

CAUSE: Pressure release

ACTIVITY: Construction, commissioning, decommissioning

RULE: Bypassing safety controls

NARRATIVE:

The silencer internals failed, resulting in debris being ejected up to 35m. At the time of the incident, 15 authorized people were within this radius. The closest being 3 valve operators who were standing 22m from the silencer vent. No one was injured from the falling materials.

WHAT WENT WRONG:

- Silencer not being fabricated to engineering design resulted in a catastrophic failure of the internal parts. During fabrication at site, weaker construction materials were used.
- Presence of a welding blanket within the system pipework helped propagate the internals failure.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

- Are your temporary equipment processes as robust as your permanent processes?
- How critical is your line cleaning in relation to subsequent commissioning activities?
- How do you minimize or eliminate non-essential personnel from being within controlled areas during higher risk tasks?

CAUSAL FACTORS:

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate design/specification/management of change

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate/defective tools/equipment/materials/products

DATE: 13 Aug 2022

COUNTRY: Indonesia

FUNCTION: Construction

CAUSE: Pressure release

ACTIVITY: Construction, commissioning, decommissioning

RULE: Line of fire

NARRATIVE:

A worker was struck by pipe movement while operating a manual valve which was being closed to build-up pressure and then opened to clear the piping system of any debris. The incident occurred when the pressure release rapidly causing the spool to shift suddenly, striking the worker.

WHAT WENT WRONG:

Pre-task risk assessment did not recognize hazard risks of:

- line of fire from air blowing moving the pipe and entrapping the worker between the pipe and a fixed point.
- the control valve was located behind the operator.

ASIA/AUSTRALASIA ONSHORE

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

Incident prevention starts by designing barriers into the work activities. Pressurized air blowing activities should consider the use of remotely operated valves. Pre-task assessments should include verification of work location and that the workers understand the risks of the task.

CAUSAL FACTORS:

PEOPLE (ACTS): Following Procedures: Improper position (in the line of fire)

PEOPLE (ACTS): Use of Protective Methods: Equipment or materials not secured

DATE: 13 Aug 2022

COUNTRY: Malaysia

FUNCTION: Production

CAUSE: Explosion, fire or burns

ACTIVITY: Maintenance, inspection, testing

RULE: Other

NARRATIVE:

The heat up preparation was started. One of the activities performed was the removal of the heat up burner (HUB) plug. After the removal of the plug and civil inspection, flash fire incident occurred while waiting for the installation of the heat up burner. One of the contractor partners was caught in the flash fire and suffered a burn injury.

WHAT WENT WRONG:

Most likely root cause of the flash fire is the combination of 3 factors.

1. Hydrocarbon presence due to 10UZ412 3%-5% opening or possible passing valve.
2. Breaking of the flange under vacuum conditions introducing atmospheric air into the reactor.
3. Temperature in the reactor still above autoignition temperature (>500 degrees C).

Process Safety Fundamental: We maintain safe isolation. We walk the line.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

- Review the current practice of isolating the 10UZX11 upstream block valve after equipment startup as it has the potential to send hydrocarbon to reactors if the 10UZX12 valve is passing, propose the best way forward.
- Review the existing heat up burner design for a safer design.
- Review the requirement of field operator to be at site all the time monitoring/supervising maintenance, turnaround and construction during HUB removal/installation activity.

CAUSAL FACTORS:

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate design/specification/management of change

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate/defective tools/equipment/materials/products

PROCESS (CONDITIONS): Organizational: Inadequate hazard identification or risk assessment

ASIA/AUSTRALASIA ONSHORE

DATE: 17 May 2022

COUNTRY: New Zealand

FUNCTION: Unspecified

CAUSE: Exposure electrical

ACTIVITY: Maintenance, inspection, testing

RULE: Energy isolation

NARRATIVE:

On Tuesday the 17th of May at 3.45pm, electrical work was being undertaken in substation one. A person was involved in fitting an air circuit breaker when they inadvertently contacted live equipment.

WHAT WENT WRONG:

No permit to work.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

- Only carry out work that is within your permit scope/isolation procedure.
- If the scope changes or alternative work is proposed return to the permit issuer to discuss/amend the permit/isolation certificate.
- Be aware that the status of the plant/equipment can change from day to day.
- Exposed, live electrical parts must be de-energized before work on or near them is permitted. Prevent accidents and isolate electrical energy by locking and tagging out the electrical system or parts of the system according to your company's lock out tag out policy.
- Prove the isolation before starting work.
- Ensure you are wearing the correct PPE for the work being undertaken.

CAUSAL FACTORS:

PEOPLE (ACTS): Inattention/Lack of Awareness: Improper decision making or lack of judgment

PROCESS (CONDITIONS): Organizational: Inadequate hazard identification or risk assessment

DATE: 16 Mar 2022

COUNTRY: Papua New Guinea

FUNCTION: Unspecified

CAUSE: Dropped objects

ACTIVITY: Transport - Air

RULE: Line of fire

NARRATIVE:

On March 16th, a M-8 Helicopter was about to lift a cargo net of 10x200l Fuel Drums at the Base laydown. The load master had connected the cargo net lifting points of the chains, and signalled for the aircraft to lift.

As the longline started to become taut the loadmaster started to move away to the safe area and noticed the lifting points of the net dropped with the weight of the chains. When he looked up, he observed the longline falling down in a spiral loop from the aircraft and moved further away to a safe location. The flight engineer advised the captain flying that the hook had released. The captain checked with the load master if he was safe and after confirmation the aircraft returned to the helipad.

ASIA/AUSTRALASIA ONSHORE

The 52m longline with load cell (75kg total) came to stop on the ground approximately 7 metres from the load and 6 metres away from the loadmaster.

No personnel were injured, and no equipment was damaged.

WHAT WENT WRONG:

- Presence of moisture in the connector resulted in the arcing between the Pin 11 and Pin 16 creating a circuit causing the emergency hook release of the load.
- Operating in high humidity and high rain fall environment.
- Close proximity of pin 11 to pin 16.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

- Ensure the maintenance program for parked machines is implemented with checks on critical components.
- Inspect the multiple pin connectors for wet and corrosion tracks during the manufacturers specified 100 hourly service.
- Inspect the hook control line for correct DC power (27V) during the manufacturers specified 100 hourly service.
- Use dehumidifier in aircraft operating in high humidity and rain fall environment when parked.
- Use covers/tarps over cockpit during heavy rain events.
- Ensure that escape routes for the loadmasters are identified and free of obstacles.
- Ensure that the laydown area is free of non-essential personnel.

CAUSAL FACTORS:

PEOPLE (ACTS): Following Procedures: Improper lifting or loading

PEOPLE (ACTS): Use of Tools, Equipment, Materials and Products: Improper use/position of tools/equipment/materials/products

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate design/specification/management of change

DATE: 10 Oct 2022

COUNTRY: Thailand

FUNCTION: Drilling

CAUSE: Dropped objects

ACTIVITY: Office, warehouse, accommodation, catering

RULE: Line of fire

NARRATIVE:

The tree fell during thunderstorm and broke the window glass and damaged the fence.

WHAT WENT WRONG:

The Main Camp experienced a heavy thunderstorm. The tree outside the location fell across the road.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

Lack of risk assessment of nature impact.

CAUSAL FACTORS:

PROCESS (CONDITIONS): Work Place Hazards: Storms or acts of nature

ASIA/AUSTRALASIA ONSHORE

DATE: 11 Jul 2022

COUNTRY: Thailand

FUNCTION: Production

CAUSE: Falls from height

ACTIVITY: Production operations

RULE: Other

NARRATIVE:

An operator performed a petrol site check around the well head then fell into a well cellar.

WHAT WENT WRONG:

The operator walked on a grating over a well cellar. 10:43 The grating turned over and the Operator fell down. 10:44 The security guard heard a loud noise, so he went down into the cellar and supported him up to the ground and took to safe area. 10:46 The Operator informed CRO and Shift supervisor. 10:51 The pick-up P1 took the injured person to the local hospital. 11:15 Arrived local hospital.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

Have close supervision and communicate with the team to improve team communication and job planning.

CAUSAL FACTORS:

PROCESS (CONDITIONS): Organizational: Inadequate hazard identification or risk assessment

PROCESS (CONDITIONS): Organizational: Inadequate communication

ASIA/AUSTRALASIA OFFSHORE

DATE: 11 Jul 2022

COUNTRY: Australia

FUNCTION: Drilling

CAUSE: Falls from height

ACTIVITY: Transport - Water, incl. marine activity

RULE: Bypassing safety controls

NARRATIVE:

A marine crew member (IP) was attempting to untangle a bundle of wire rope slings on board an anchor handling vessel. In doing so, the IP stepped backwards and fell into an open chain locker. The chain locker is approximately 7 metres deep, the IP was able to arrest his fall by maintaining his grip on the wire rope sling that he was handling. His grip minimised the impact when he landed at the bottom of the empty chain locker.

WHAT WENT WRONG:

- Procedure not followed – use of temporary safety barriers was not implemented after the removal of gratings.
- Situational Awareness - IP miscalculated the location of the open chain locker or otherwise mistook the open chain locker as normal decking.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

- Ensure risk assessments are reviewed and updated when conditions change or new hazards are introduced.
- Clarity on roles and responsibilities when removing barriers/guards that create a fall from height risk.
- Safety harness to be worn when there is a potential fall from height risk.
- Use hard barricading to prevent falls when guarding/barriers are removed.

CAUSAL FACTORS:

PEOPLE (ACTS): Use of Protective Methods: Inadequate use of safety systems

PEOPLE (ACTS): Use of Protective Methods: Personal Protective Equipment not used or used improperly

PROCESS (CONDITIONS): Protective Systems: Inadequate/defective guards or protective barriers

PROCESS (CONDITIONS): Organizational: Inadequate hazard identification or risk assessment

PROCESS (CONDITIONS): Organizational: Inadequate supervision

DATE: 15 Oct 2022

COUNTRY: Indonesia

FUNCTION: Drilling

CAUSE: Dropped objects

ACTIVITY: Drilling, workover, well operations

RULE: Safe mechanical lifting

NARRATIVE:

Uncontrolled descent of mudlogging unit while being moved by rig crane.

WHAT WENT WRONG:

Crane was overloaded and fly hoist motor failed.

ASIA/AUSTRALASIA OFFSHORE

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

- Inadequate risk assessment.
- Improved control of crane overrides required.

CAUSAL FACTORS:

PEOPLE (ACTS): Following Procedures: Improper lifting or loading

PEOPLE (ACTS): Use of Protective Methods: Inadequate use of safety systems

PEOPLE (ACTS): Inattention/Lack of Awareness: Improper decision making or lack of judgment

DATE: 18 Jan 2022

COUNTRY: Malaysia

FUNCTION: Production

CAUSE: Struck by (not dropped object)

ACTIVITY: Production operations

RULE: Work authorization

NARRATIVE:

While the personnel were trying to rectify a flow control valve (FCV) that has been hunting, the team identified that the control valve issue was due to the actuator internals soft kit having been worn out.

While trying to rectify the actuator by unbolting of the spring retainer, the bolt, spring plate and spring were ejected vertically (under tension), hitting a pipe before landing on the floor grating.

No one was injured in the incident as all personnel were consciously avoiding being in the line of fire.

WHAT WENT WRONG:

The Team Lead proceeded with the disassembly of the actuator, while reasoning that the permit used for the troubleshooting was adequate to manage the associated risk of opening the actuator to adjust the positioning of the internal soft kit.

2 of the work crew members attempted to intervene and stop the work due to the potential stored energy in the actuator, but this was not taken into account by the Team Lead.

All crew members stayed clear of the line of fire.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

- Importance of complying with Safe Work Practices and Requirements, e.g. Work Authorization & PTW, and being able to raise concerns if an individual may not be able to follow the rule.
- Apply effective intervention and stop work where there is an immediate threat to the safety and health of the employees or adverse impact to the environment.
- Create Psychologically Safety & Respectful Environment; for all to feel safe to speak and share ideas/concerns.

CAUSAL FACTORS:

PEOPLE (ACTS): Following Procedures: Deviation intentional (by individual or group)

PEOPLE (ACTS): Use of Tools, Equipment, Materials and Products: Improper use/position of tools/equipment/materials/products

PEOPLE (ACTS): Inattention/Lack of Awareness: Improper decision making or lack of judgment

ASIA/AUSTRALASIA OFFSHORE

DATE: 22 Dec 2022

COUNTRY: Myanmar

FUNCTION: Construction

CAUSE: Dropped objects

ACTIVITY: Lifting, crane, rigging, deck operations

RULE: Bypassing safety controls

NARRATIVE:

Dropped objects from Telescopic Boom Crane.

WHAT WENT WRONG:

The crane operator started the engine of a telescopic crane for warming up prior to performing the operation. Suddenly he saw two of the slippers had fallen onto the portside mezzanine deck. The Crane Operator stopped the crane and checked. It was observed that the slippers had fallen from the crane boom to the portside mezzanine deck area close to the survey pool tugger.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

Improve the preventive maintenance system on-board the telescopic crane.

CAUSAL FACTORS:

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate design/specification/management of change

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate/defective tools/equipment/materials/products

DATE: 21 Aug 2022

COUNTRY: New Zealand

FUNCTION: Production

CAUSE: Dropped objects

ACTIVITY: Maintenance, inspection, testing

RULE: Working at height

NARRATIVE:

At approx. 11:00am, the IP was preparing to carry out derrick inspection. After hooking onto the cable sleeve the IP stepped on the bottom rung of the derrick ladder and put his body weight onto the cable sleeve to test it. This resulted in the wire coming free from the top socket and the IP fell backwards onto the Calvin Slip Bowl which was lying 1.2m west of the bottom of the ladder.

WHAT WENT WRONG:

- Incorrectly installed equipment: the cable did not protrude 25-50mm above the cable clamp.
- Cable and fixing assembly were corroded.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

- If you have cable sleeve units with this style of upper bracket on your facility, please remove from service until they can be inspected and deemed fit for purpose.
- Note they are unable to be inspected without disassembly allowing the wire and carrier clamp assembly to be removed from the top bracket pipe.

ASIA/AUSTRALASIA OFFSHORE

- It should also be noted that the manufacturers have modified the design on the upper bracket on more recent models.
- This new design can be inspected without disassembly and should not be affected by the issues described above.

CAUSAL FACTORS:

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate/defective tools/equipment/materials/products

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate maintenance/inspection/testing

DATE: 10 Nov 2022

COUNTRY: Philippines

FUNCTION: Production

CAUSE: Dropped objects

ACTIVITY: Maintenance, inspection, testing

RULE: Bypassing safety controls

NARRATIVE:

Two pieces of detached ceiling insulation cladding were found on top of a pipe. These claddings were used to hold ceiling insulation. One stainless steel sheet has a weight of 11 kg and another angular stainless-steel cladding has a weight of 13 kg. Dropped height is at around 7-8 metres, which, with the weight of the dropped objects, will qualify for a potential fatality consequence as per the DROPS calculator.

WHAT WENT WRONG:

Dropped Objects, Corrosion and Aging Asset.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

Consider reviewing potential dropped objects especially those that do not have preventive maintenance routines.

CAUSAL FACTORS:

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate maintenance/inspection/testing

DATE: 26 Feb 2022

COUNTRY: Thailand

FUNCTION: Drilling

CAUSE: Struck by (not dropped object)

ACTIVITY: Transport - Water, incl. marine activity

RULE: Line of fire

NARRATIVE:

At approximately 16:00 hrs, the operation was stand by on the weather at the shelter area. A Bosun, an Able Seaman (AB) and a cadet were transferring equipment from the ship's office to the ship's garage under the crane along the ship's safe walkway.

They observed that the ladder that had previously been secured to the ship's side railing started to become loose and they went out of the safe zone walkway to the side railing to re-secure the steps. At this time there were heavy swells coming past the vessel but they did not observe any coming over the side as they watched for a few minutes. While they started to re-secure the steps and had taken their eyes off the incoming sea, a large swell around 3 – 4m

ASIA/AUSTRALASIA OFFSHORE

came over the side of the bulwarks of the vessel and hit the ABs (IP). The swell was strong enough for him to lose his footing and he fell on the deck and was pushed down the deck to just forward of the crane pedestal.

The Bosun and Cadet came to assist the IP who was unable to get back up due to an injury to his ankles. The IP was carried by stretcher to the sick bay where the IP was assessed by the medic.

The Incident happened very quickly, and there is no evidence to show exactly what has caused the injuries to the IP.

WHAT WENT WRONG:

Pressure from swell knocked IP down.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

Not available.

CAUSAL FACTORS:

PEOPLE (ACTS): Use of Protective Methods: Equipment or materials not secured

PEOPLE (ACTS): Inattention/Lack of Awareness: Lack of attention/distracted by other concerns/stress

DATE: 18 May 2022

COUNTRY: Thailand

FUNCTION: Production

CAUSE: Dropped objects

ACTIVITY: Lifting, crane, rigging, deck operations

RULE: Line of fire

NARRATIVE:

Wooden bar support crane boom rest dropped onto platform.

WHAT WENT WRONG:

On 18 May 2022 at 0700 hrs., after completing crew change activity, while a Production Operator (holding valid crane license class B) was controlling a crane boom to park onto crane boom rest. The wooden bar on the top of the crane boom rest suddenly dropped onto the platform where a rigger was working nearby, approximately 5 metres away from the boom rest.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

Material used did not anticipate moisture accumulated conditions. Design did not anticipate inspection program.

CAUSAL FACTORS:

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate design/specification/management of change

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate maintenance/inspection/testing

ASIA/AUSTRALASIA OFFSHORE

DATE: 25 Apr 2022

COUNTRY: Vietnam

FUNCTION: Production

CAUSE: Unspecified - Other

ACTIVITY: Production operations

RULE: Unspecified

NARRATIVE:

Gas release from gas lift compressor intercooler.

WHAT WENT WRONG:

A tube bundle in the gas lift compressor inter-cooler burst and 453kg of hydrocarbon gas was released to the atmosphere.

Process Safety Fundamental: Not applicable.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

Poor documentation of material selection during design led to lack of consideration for external corrosion as a failure mechanism for the cooling system.

CAUSAL FACTORS:

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate design/specification/management of change

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate/defective tools/equipment/materials/products

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate maintenance/inspection/testing

EUROPE ONSHORE

DATE: 29 Jan 2022

COUNTRY: Norway

FUNCTION: Production

CAUSE: Dropped objects

ACTIVITY: Construction, commissioning, decommissioning

RULE: Line of fire

NARRATIVE:

During strong winds three scaffolding planks fell to the ground, a drop of about 23 metres, with an expended energy of 1800 Joules. No one was hit, but there were people in the area.

WHAT WENT WRONG:

- Shelter/tent construction was not strong enough to withstand high winds.
- Missing requirements for following up and inspecting temporary constructions.
- Missing requirements for securing site in extreme weather conditions.
- Decision to close site due to weather was not made in time (no instructions or set criteria).
- Unclear divisions of responsibility between disciplines and departments.
- Shift leader work responsibilities not formalised and communicated.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

- Check the local forecast and plan all operation in according to the weather.
- Discuss the weather conditions with your teammates.
- Secure all loose items.

CAUSAL FACTORS:

PROCESS (CONDITIONS): Organizational: Inadequate work standards/procedures

PROCESS (CONDITIONS): Organizational: Inadequate communication

PROCESS (CONDITIONS): Organizational: Poor leadership/organizational culture

DATE: 09 Feb 2022

COUNTRY: Norway

FUNCTION: Production

CAUSE: Dropped objects

ACTIVITY: Construction, commissioning, decommissioning

RULE: Line of fire

NARRATIVE:

During inspection, one of the participants bumped into a loose plank stored on the scaffolding. The plank fell into a manhole to the level below. The area was not cordoned off. There were no personnel in the area below.

WHAT WENT WRONG:

Person bumped into loose plank. Area below not cordoned off.

EUROPE ONSHORE

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

- Always have control of the workplace with regards to loose items, especially when working at height.
- Secure loose items before start-up.
- Remember to notify all incidents immediately.

CAUSAL FACTORS:

PROCESS (CONDITIONS): Organizational: Inadequate hazard identification or risk assessment

PROCESS (CONDITIONS): Organizational: Inadequate supervision

DATE: 12 Feb 2022

COUNTRY: Norway

FUNCTION: Production

CAUSE: Dropped objects

ACTIVITY: Construction, commissioning, decommissioning

RULE: Line of fire

NARRATIVE:

During installation of the green sea panel by ship side, a steel wedge came out of position and fell down and hit the railing on the quay, bounced and landed on some tarp between the boat and the quay.

WHAT WENT WRONG:

- Wedge came out of position and fell down.
- The wedge was not secured.
- Blue net was not secured again after removing water.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

- Review if safety nets should be permitted to open by others than scaffold department and how it could be possible to prevent work from been commenced before barrier was re-established.
- Review the quality of their input process to JSA and the competence of crew to obtain a knowledge and skill necessary to document concrete mitigating actions and detail in the documents.

CAUSAL FACTORS:

PROCESS (CONDITIONS): Organizational: Inadequate hazard identification or risk assessment

DATE: 31 Oct 2022

COUNTRY: Norway

FUNCTION: Production

CAUSE: Dropped objects

ACTIVITY: Maintenance, inspection, testing

RULE: Line of fire

NARRATIVE:

Loose cable tray was unsecured on top of existing cable tray. The loose tray fell and landed on gangway. The area was not cordoned off.

EUROPE ONSHORE

WHAT WENT WRONG:

- Ensure good housekeeping.
- Risk assess storing of equipment.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

- Ensure good housekeeping.
- Risk assess storing of equipment.

CAUSAL FACTORS:

PEOPLE (ACTS): Inattention/Lack of Awareness: Fatigue

PROCESS (CONDITIONS): Organizational: Inadequate hazard identification or risk assessment

PROCESS (CONDITIONS): Organizational: Inadequate supervision

DATE: 01 Feb 2022

COUNTRY: Norway

FUNCTION: Construction

CAUSE: Dropped objects

ACTIVITY: Lifting, crane, rigging, deck operations

RULE: Working at height

NARRATIVE:

A 2.35 m long rail (2.55 kg) for fixing of tarpaulin fell from the underside of the main deck in connection with dismantling of the emergency stair tower at a column by a team of hired scaffolders.

The rail hit the scaffolding floor above the pontoon deck approximately 24.5 m below. The pontoon deck was cordoned off, but there were three scaffolders from the dismantling team inside the barriers without paying attention to the work directly above.

The rail hit the scaffolding floor less than 0.5 m from one of them, where it made a dent in the scaffolding plank.

WHAT WENT WRONG:

Insufficient leadership:

Managers had not taken leadership of the work to secure safe execution:

- Work at height has been performed without securing loose objects.
- Work at height without ensuring that there are no personnel in the potential Drop Zone.
- Lack of communication between two work parties.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

When doing Work at height, always ensure that the potential drop zone is free.

- Objects must always be secured during work at height.
- Perform a risk assessment or a Toolbox talk before working at height.
- When working at height, it must be clear for all involved how the work inside the barricaded area is organized and a means of communication must be available.

EUROPE ONSHORE

CAUSAL FACTORS:

PEOPLE (ACTS): Use of Protective Methods: Failure to warn of hazard

PEOPLE (ACTS): Use of Protective Methods: Equipment or materials not secured

PEOPLE (ACTS): Inattention/Lack of Awareness: Improper decision making or lack of judgment

PROCESS (CONDITIONS): Organizational: Inadequate communication

PROCESS (CONDITIONS): Organizational: Inadequate supervision

DATE: 06 Oct 2022

COUNTRY: Romania

FUNCTION: Drilling

CAUSE: Dropped objects

ACTIVITY: Drilling, workover, well operations

RULE: Safe mechanical lifting

NARRATIVE:

While L/D directional drilling BHA, after multi frequency resistivity (MFR) tool was pulled out from well, this tool was secured on one end with winch (wire sling) and on the other end with the crane (wire sling). During L/D operation/ moving manoeuvres from the rig floor to the catwalk the tool slipped from the winch and remained hanging just on crane sling. The result was that it fell to the ground with one end close to the catwalk. The directional drilling tool and two casing joints (from the ramp) were damaged.

WHAT WENT WRONG:

- Use of inappropriate lifting/handling devices.
- Incorrect positioning of the crane banksman.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

- Revision of handling/lifting procedures for contractors' tubular equipment.
- Verification of procedures for handling/lifting tubular material, equipment, drilling contractors.
- Revision of JSA procedures for handling/lifting tubular material.

CAUSAL FACTORS:

PROCESS (CONDITIONS): Organizational: Inadequate work standards/procedures

PROCESS (CONDITIONS): Organizational: Inadequate hazard identification or risk assessment

PROCESS (CONDITIONS): Organizational: Inadequate supervision

DATE: 14 Dec 2022

COUNTRY: Romania

FUNCTION: Production

CAUSE: Explosion, fire or burns

ACTIVITY: Production operations

RULE: Hot work

NARRATIVE:

During welding works at a pipeline located near a saltwater tank, an explosion followed by fire occurred at the saltwater tank. The roof of the saltwater tank was thrown away, damaging a nearby oil tank. Several barracks and equipment were damaged. Also, some third-party buildings in the vicinity were damaged.

EUROPE ONSHORE

WHAT WENT WRONG:

Lack of accountability for site ownership responsibility in terms of safe operations practices

- Non-observance of work task regarding gas testing before and during the works (according to Hot Works Permit).
- Failure to establish and to observe the proper safety measures inside the facility upon verification and approval of the Hot Works Permit.
- No training organized on the operator PTW System (JSA, SIMOPS, Isolation).
- Lack of risk awareness on the hydrocarbon gas hazards from executant and operators.
- Operators poor training on process safety risks.
- Lack of communication between contractor companies working.
- Stop work authority not applied by the employees of all companies present on site.
- Missing competency of coordinator of pipe execution team and execution team lack of competence on the Contractor safe system of work (PTW system).
- The mechanical & process isolation procedure was not applied.

Process Safety Fundamental: We control ignition sources.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

- Common Process and Integrity risk assessment for all facilities operated by contractor.
- Joint evaluation of operational documents (e.g., Operations Manual, P&IDs) to identify and update the documents to reflect current plant conditions, as built status and foreseeable normal, abnormal and emergency situations. Train operators on content of Operations Manual (normal, abnormal, emergency situations). Evaluate the extension of review to other facilities operated by contractor.
- Develop an HSE Procedure regarding contractors for projects within operated facilities – 3 parties' protocol (company/contractor/external contractors); A hand over protocol should be in place for each work performed in those facilities – access on facility, training campaign, HSE responsibility, etc.

CAUSAL FACTORS:

PEOPLE (ACTS): Following Procedures: Deviation unintentional (by individual or group)

PEOPLE (ACTS): Use of Protective Methods: Failure to warn of hazard

PEOPLE (ACTS): Use of Protective Methods: Inadequate use of safety systems

PEOPLE (ACTS): Use of Protective Methods: Equipment or materials not secured

PEOPLE (ACTS): Inattention/Lack of Awareness: Improper decision making or lack of judgment

PROCESS (CONDITIONS): Organizational: Inadequate training/competence

PROCESS (CONDITIONS): Organizational: Inadequate hazard identification or risk assessment

PROCESS (CONDITIONS): Organizational: Inadequate communication

EUROPE OFFSHORE

DATE: 17 Jun 2022

COUNTRY: Denmark

FUNCTION: Construction

CAUSE: Pressure release

ACTIVITY: Construction, commissioning, decommissioning

RULE: Bypassing safety controls

NARRATIVE:

An operations team were preparing a 14" pipeline riser for reconnection to a new topside facility service with the cap suitable for the associated axial load.

The 14" condensate export line had remained isolated, water-filled, inhibited, and decommissioned since the project hook down phase in 2019. The line was positively isolated at one end and fitted with an engineered cap at the other end. The cap was connected to the riser via a pipe coupling. Both cap and coupling were rated for 10bar service with the cap suitable for the associated axial load. As 100% lower exposure limit was identified when sampling the void space contents at one end, as per local procedure, the team were attempting to perform Nitrogen pressure purges of the 10-metre void space to displace residual hydrocarbon before removal of the cap to allow welding of a connecting pup-spool to the open riser. Whilst trying to locally verify the applied N2 purge pressure, the engineered cap and pipe coupling were vertically ejected. The equipment and debris resulted in physical injuries to three people in the vicinity; Injured Person 1 was lifted upward and fell back to scaffold, Injured persons 2 and 3 suffered minor facial injuries from impact with debris and/or equipment.

WHAT WENT WRONG:

- Applied N2 purge pressure exceeded axial strength of pipe coupling.
 - Improper installation of coupling during hook down, invalidated pressure rating of assembly.
- Uncertainty of pressure applied due to use of ineffective purge equipment (wrong pressure indicator range, no overpressure protection, inadequate control of pressure).
- Multiple workers in vicinity of work site.
 - Access to work site not controlled (no barriers).
 - Failure to mitigate Line of Fire hazard (N2 manifold locally mounted at riser location).
- Purging proceeded without stopping job.
 - Recognized concerns did not lead to an effective intervention.
- Inadequate Pipeline/Process integrity assurance process.
 - Installation of 10bar coupling during hook down did not lead to formal derating of pipeline.
 - Poor installation and lack of 'mechanical completion' checks at hook down
 - Operational hook-up needs not fully considered at hook down phase.

Process Safety Fundamental: We apply procedures. We stop if the unexpected occurs.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

- Ensure that process/pipeline integrity is maintained for temporarily decommissioned lines and equipment (derating, testing, verification, recording and assurance).
- Ensure that site procedures provide adequate mitigation of risks associated with N2 purging activities, and that overpressure protection, system integrity check, and correct means of pressure control/monitoring are assured as prerequisite controls.

EUROPE OFFSHORE

- Ensure that purge and/or pressure test equipment is located to mitigate line of fire hazard.
- Ensure that pressure purging activities are regarded as high-risk, and systematically subjected to method statement, risk assessment and PTW.

CAUSAL FACTORS:

PEOPLE (ACTS): Use of Tools, Equipment, Materials and Products: Improper use/position of tools/equipment/materials/products

PROCESS (CONDITIONS): Protective Systems: Inadequate/defective guards or protective barriers

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate design/specification/management of change

DATE: 17 Jan 2022

COUNTRY: Norway

FUNCTION: Drilling

CAUSE: Dropped objects

ACTIVITY: Drilling, workover, well operations

RULE: Safe mechanical lifting

NARRATIVE:

When pulling a 13 3/8" Intercept plug (length 9.9 m, weight 2.8 tons) the plug dislodged from the running tool as the tugger wire attached to the lower end started to take load to initiate tailing out the assembly. The lower point of the assembly was at this point approx. 0.5 m above the rig floor. When the plug impacted on the rig floor, it parted and broke into pieces.

WHAT WENT WRONG:

The importance of checking the overshot and lug positioner sleeve was not covered in the procedure.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

Update procedure to include checking of the overshot and lug positioner sleeve position when retrieved from the wellbore.

Training to be provided to operations personnel to ensure familiarization and competence relating to handling of Intercept plug.

CAUSAL FACTORS:

PEOPLE (ACTS): Following Procedures: Improper lifting or loading

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate maintenance/inspection/testing

PROCESS (CONDITIONS): Organizational: Inadequate training/competence

PROCESS (CONDITIONS): Organizational: Inadequate communication

EUROPE OFFSHORE

DATE: 20 Dec 2022

COUNTRY: Norway

FUNCTION: Drilling

CAUSE: Dropped objects

ACTIVITY: Drilling, workover, well operations

RULE: Working at height

NARRATIVE:

After skidding, the drilling rig part of a railing, 1m wide, was to be reinstalled. During installation, the handrail fell to the deck below.

Weight of railing part: 35.7 kg.

Fall height: 21.5 m.

WHAT WENT WRONG:

The railing was not secured during reinstallation, the only barrier preventing it from falling was the person holding it. High winds, 23 knots, 27 knots gusts contributed to the incident.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

Revise procedure to include the requirement of additional securing measures while installing/reinstalling railing parts.

CAUSAL FACTORS:

PEOPLE (ACTS): Use of Protective Methods: Inadequate use of safety systems

PROCESS (CONDITIONS): Protective Systems: Inadequate security provisions or systems

PROCESS (CONDITIONS): Organizational: Inadequate work standards/procedures

PROCESS (CONDITIONS): Organizational: Inadequate hazard identification or risk assessment

DATE: 25 May 2022

COUNTRY: Norway

FUNCTION: Drilling

CAUSE: Dropped objects

ACTIVITY: Lifting, crane, rigging, deck operations

RULE: Safe mechanical lifting

NARRATIVE:

During backloading of cargo part of a gate on the crane's ring platform fell to the deck below. The port had been fastened using a ratchet strap. A loose end had unintentionally been tangled in the ladder as the crane turned around.

Weight of gate part: 1.24 kg.

Fall height: 15.7 m.

EUROPE OFFSHORE

WHAT WENT WRONG:

- Improper design of the securing of the gate.
- Inadequate risk assessment and inspection.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

Ensure proper hazard identification/risk assessment prior to crane operations, including inspection of gates at the crane's ring platform.

Minimize the use of ratchet straps for securing of gates. A more permanent solution to be installed.

CAUSAL FACTORS:

PEOPLE (ACTS): Following Procedures: Improper lifting or loading

PEOPLE (ACTS): Use of Protective Methods: Inadequate use of safety systems

PROCESS (CONDITIONS): Protective Systems: Inadequate security provisions or systems

PROCESS (CONDITIONS): Organizational: Inadequate work standards/procedures

PROCESS (CONDITIONS): Organizational: Inadequate hazard identification or risk assessment

DATE: 17 Dec 2022

COUNTRY: Norway

FUNCTION: Drilling

CAUSE: Falls from height

ACTIVITY: Lifting, crane, rigging, deck operations

RULE: Working at height

NARRATIVE:

Rope access climbers were working on a deck rigging up firewater hose for fire monitors. To connect the hose, they needed one more chain hoist to be connected up in order to lift/support the hose while connecting it to the flange. One of the persons in the team connected a webbing sling to the chain hoist and prepared to choke the sling around the deck support beam. The person realized he was not reaching high enough to get the webbing sling around the beam, so he took one step up on the lowest horizontal pipe on the handrail. To stabilize himself, he then took his other foot over the handrail. At this point he was per definition working over the side without any harness.

WHAT WENT WRONG:

The person did not use fall arrest equipment due to underestimation of the risk involved.

When to use fall arrest equipment during the work at hand had not been predefined.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

- Ensure toolbox talks/safe job analyses sufficiently cover the work tasks at hand and the associated risks.
- Rope access climbers to use fall arrest equipment when working at Texas deck.

CAUSAL FACTORS:

PEOPLE (ACTS): Use of Protective Methods: Inadequate use of safety systems

PEOPLE (ACTS): Use of Protective Methods: Personal Protective Equipment not used or used improperly

PEOPLE (ACTS): Inattention/Lack of Awareness: Improper decision making or lack of judgment

EUROPE OFFSHORE

DATE: 17 Oct 2022

COUNTRY: Norway

FUNCTION: Production

CAUSE: Caught in, under or between (excl. dropped objects)

ACTIVITY: Maintenance, inspection, testing

RULE: Energy isolation

NARRATIVE:

The oil export pump shaft started to rotate while personnel was working up close to the shaft. The work, adjusting a sensor, had just ended when the shaft started to turn. The rotation of the shaft was caused by the starting of the upstream oil booster pump. This caused the fluid in the line from the booster pump to turn the impeller on the export pump B, causing the shaft to spin. This could happen as there was no mechanical isolation of the oil export pump.

WHAT WENT WRONG:

- There was no mechanical isolation of the oil export pump.
- Lack of coordination of simultaneous activities.
- The description of the job was added to an approved PTW.
- Perceived time shortage.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

Improve the work permit (WP) process based on lessons learned from projects that have used it.

- Make sure that the greenfield WP process is improved based on experiences from all projects that have used the process.
- Bring in more focus on mechanical isolation for work on rotating equipment. Define requirements to mechanical isolation according to requirements to electrical isolation.
- Set stricter requirements on how to use a general WP, including defining a general WP.
- Set requirements to use the brownfield WP process for pressurized and energized mechanical systems, not only when HC is present.
- When the scope is changed after WP has been approved, the WP should be reapproved. This should be considered as a point/requirement in the WP process.
- Improve quality in WP descriptions and be compliant to the requirements. Description and attachment to the WP. It shall be possible to understand the task and the corresponding risk to each task in a WP.

CAUSAL FACTORS:

PROCESS (CONDITIONS): Protective Systems: Inadequate/defective Personal Protective Equipment

PROCESS (CONDITIONS): Protective Systems: Inadequate/defective warning systems/safety devices

PROCESS (CONDITIONS): Organizational: Inadequate work standards/procedures

PROCESS (CONDITIONS): Organizational: Inadequate communication

PROCESS (CONDITIONS): Organizational: Inadequate supervision

EUROPE OFFSHORE

DATE: 16 Jan 2022

COUNTRY: Norway

FUNCTION: Production

CAUSE: Dropped objects

ACTIVITY: Drilling, workover, well operations

RULE: Line of fire

NARRATIVE:

During loading bulk barite on starboard side loading station from supply boat, a lid on the Rock Catcher in the bulk line came loose and landed FWD on the lay down area.

WHAT WENT WRONG:

Difficult to open clamp on starboard barite rock catcher as the handle for the Victaulic clamp on starboard side was hard to open. The assistant driller had first to use hammer, then he used the crowbar, not optimal solution for regular clean out of rock catchers.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

Align Maintenance PMs on the Rock Catchers.

Knowledge and awareness towards Governing Documents.

Install secondary retention on the starboard / Port rock catchers.

CAUSAL FACTORS:

PEOPLE (ACTS): Following Procedures: Deviation unintentional (by individual or group)

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate maintenance/inspection/testing

PROCESS (CONDITIONS): Organizational: Inadequate work standards/procedures

PROCESS (CONDITIONS): Organizational: Inadequate supervision

DATE: 28 Oct 2022

COUNTRY: Norway

FUNCTION: Production

CAUSE: Dropped objects

ACTIVITY: Drilling, workover, well operations

RULE: Safe mechanical lifting

NARRATIVE:

When breaking and laying down 5 ½" HWDP singles a dropped object occurred falling approximately 20 metres from the Tubular Feeding Machine (TFM). The dropped object was one of the TFM's side bumpers which was found inside Red Zone drill pipe setback.

WHAT WENT WRONG:

Dropped TFM side bumper when it got caught on wire sling when running the TFM downward to horizontal position.

EUROPE OFFSHORE

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

- Improvement of the lighting conditions and CCTV camera coverage is needed.
- Evaluate the need to have the side bumpers installed when tubulars are laying in trolley.
- Check other equipment if non-original securing devices are in use at height.

CAUSAL FACTORS:

PEOPLE (ACTS): Use of Protective Methods: Disabled or removed guards, warning systems or safety devices

DATE: 10 Nov 2022

COUNTRY: Norway

FUNCTION: Production

CAUSE: Dropped objects

ACTIVITY: Drilling, workover, well operations

RULE: Line of fire

NARRATIVE:

Crush injury.

WHAT WENT WRONG:

The IP was repairing a link on the drag chain, the drag chain fell 1/2metre and the IP was caught underneath.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

Provide adequate training to all the safety coaches, leadership, and rig team to assure process is overall effective.

CAUSAL FACTORS:

PEOPLE (ACTS): Following Procedures: Improper position (in the line of fire)

DATE: 08 Sep 2022

COUNTRY: Norway

FUNCTION: Production

CAUSE: Dropped objects

ACTIVITY: Lifting, crane, rigging, deck operations

RULE: Safe mechanical lifting

NARRATIVE:

A pup-joint dropped uncontrolled approximately 13,7 meters during a lifting operation. The task was to move the pup-joint from the drill floor to a storage rack outside the wind wall. At the same time a person was walking down the stairs under the lifting area. The person was not hit by the falling pup-joint or physically injured.

WHAT WENT WRONG:

During the positioning of the pup-joint, it came into contact with something and the wire slings loosened. As a result of this, the pup-joint slipped out of the wire slings and fell down to a lower level.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

- Improve training and competence management.
- Inadequate design.
- Requirements and expectations for the work must be clearly communicated and followed up.

EUROPE OFFSHORE

CAUSAL FACTORS:

PROCESS (CONDITIONS): Organizational: Inadequate training/competence

PROCESS (CONDITIONS): Organizational: Inadequate hazard identification or risk assessment

PROCESS (CONDITIONS): Organizational: Inadequate communication

PROCESS (CONDITIONS): Organizational: Inadequate supervision

DATE: 16 Nov 2022

COUNTRY: Norway

FUNCTION: Production

CAUSE: Dropped objects

ACTIVITY: Production operations

RULE: Line of fire

NARRATIVE:

The lid from a decommissioned open vent line fell on the walkway during heavy winds.

Weight of lid: 20 kg.

Fall height: 37 m.

WHAT WENT WRONG:

The vent line lid was designed to be installed on a vent facing upwards.

The lid, as installed, did not have any means of secondary retainment since the lid had not been identified as a potential dropped object.

The original tag number of the vent line was set to "deleted" after being decommissioned, hence leaving it without a formal trace in the maintenance program.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

- Review of decommissioning procedures to cover the risk of potential dropped objects.
- Review and improve inspection routines for potential dropped objects, physical and drone based.
- Review requirements for establishing tag numbers for other decommissioned equipment that could deteriorate into hazardous conditions.

CAUSAL FACTORS:

PEOPLE (ACTS): Use of Protective Methods: Equipment or materials not secured

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate maintenance/inspection/testing

PROCESS (CONDITIONS): Organizational: Inadequate work standards/procedures

PROCESS (CONDITIONS): Organizational: Inadequate hazard identification or risk assessment

EUROPE OFFSHORE

DATE: 14 May 2022

COUNTRY: Norway

FUNCTION: Production

CAUSE: Unspecified - Other

ACTIVITY: Maintenance, inspection, testing

RULE: Unspecified

NARRATIVE:

Diving operations were ongoing at vessel while lying with the wind to its side to protect the LDC (light diving craft). The FWD thruster was isolated and the turret locked in line with diving isolation matrix. The STB thruster was out of service due to an ongoing upgrade project. The Port thruster was the only thruster in operation and available.

In preparation for greasing of the port thruster steering pump 2A, the duty pump was switched from 2A to 1A. Pump 2A shutdown as per logic. Directly after, pump 1A tripped due to low-pressure alarm not clearing after start-up as per logic. The standby pump 2A did not auto-start. This resulted in no azimuth control (direction) of the thruster and loss of heading control of vessel for approx. 10 mins.

WHAT WENT WRONG:

STB pump tripped after it was transferred from standby to duty.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

- Evaluate condition on flexible connections on pumps.
- Update equipment strategy and maintenance procedure for the thruster system.
- Update and comply to SIMOPS matrix.
- Establish clear communication guidelines for diving operations.

CAUSAL FACTORS:

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate maintenance/inspection/testing

PROCESS (CONDITIONS): Organizational: Inadequate work standards/procedures

PROCESS (CONDITIONS): Organizational: Inadequate hazard identification or risk assessment

PROCESS (CONDITIONS): Organizational: Inadequate communication

DATE: 20 Jan 2022

COUNTRY: UK

FUNCTION: Production

CAUSE: Dropped objects

ACTIVITY: Lifting, crane, rigging, deck operations

RULE: Safe mechanical lifting

NARRATIVE:

Whilst executing a replacement program of constant load supports, or spring cans, a load was dropped to deck. The scaffold lifting frame erected for the task failed.

EUROPE OFFSHORE

WHAT WENT WRONG:

- The scaffold lifting frame was not built in compliance with the scaffolding contractor's technical guidance.
- Single diagonal brace provided insufficient stability/restraint against any side loading.
- The required scaffold request form used to describe its use and purpose was not completed.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

- Updated working at height procedure to ensure standardised approach to scaffold requested.
- Require lift plan to identify drop cone and equipment within.

CAUSAL FACTORS:

PEOPLE (ACTS): Following Procedures: Improper lifting or loading

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate design/specification/management of change

PROCESS (CONDITIONS): Organizational: Inadequate communication

DATE: 02 Feb 2022

COUNTRY: UK

FUNCTION: Production

CAUSE: Struck by (not dropped object)

ACTIVITY: Lifting, crane, rigging, deck operations

RULE: Safe mechanical lifting

NARRATIVE:

Contact with pressure gauge on pig receiver during lifting operations.

Container being lifted from supply vessel to a restricted landing space made contact with pressure gauge on live pig receiver. No hydrocarbon release occurred.

WHAT WENT WRONG:

- Inadequate guidance/rules for lifting adjacent to live plant.
- Inadequate risk assessment.

Process Safety Fundamentals: We respect hazards. We apply procedures. We stop if the unexpected occurs.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

- Development of deck management plan/procedure with definitive rules/guidance for lifting adjacent to live equipment.
- Additional laydown area hazard assessment to be completed.
- Standardisation of cargo management process.

CAUSAL FACTORS:

PROCESS (CONDITIONS): Work Place Hazards: Congestion, clutter or restricted motion

PROCESS (CONDITIONS): Organizational: Inadequate work standards/procedures

PROCESS (CONDITIONS): Organizational: Inadequate hazard identification or risk assessment

EUROPE OFFSHORE

DATE: 20 Jul 2022

COUNTRY: UK

FUNCTION: Construction

CAUSE: Water related, drowning

ACTIVITY: Diving (incl. decompression), subsea, ROV

RULE: Bypassing safety controls

NARRATIVE:

Unexpected excursion of dive support vessel whilst two divers were setting up local rigging. ROV was also at depth. No injuries were sustained, no equipment was damaged. Vessel Dynamic positioning (DP) system based its calculations on false position references leading to a vessel excursion.

WHAT WENT WRONG:

Alignment issues with the HiPAP transducers led to a degradation of the 2 HiPAP references within the Vessel DP System.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

- Update of masters standing DP orders to provide guidance on HPR beacon seabed placement.
- Review and update vessel operating guidance to reflect the need for three principles of reference whenever possible.
- Review of vessel checklists to ensure instruction on reference system origin and beacon placement are included.

CAUSAL FACTORS:

PEOPLE (ACTS): Following Procedures: Deviation intentional (by individual or group)

PEOPLE (ACTS): Use of Protective Methods: Disabled or removed guards, warning systems or safety devices

PEOPLE (ACTS): Inattention/Lack of Awareness: Improper decision making or lack of judgment

PROCESS (CONDITIONS): Protective Systems: Inadequate/defective guards or protective barriers

PROCESS (CONDITIONS): Organizational: Inadequate work standards/procedures

MIDDLE EAST ONSHORE

DATE: 18 Apr 2022

COUNTRY: Kurdistan Region of Iraq

FUNCTION: Drilling

CAUSE: Dropped objects

ACTIVITY: Drilling, workover, well operations

RULE: Work authorization

NARRATIVE:

While the drilling mechanical crew were on the rig floor aligning Traction Motor A with the draw-works, a jack and a half of iron roughneck roll were being used as a spacer to lift the shaft. During the alignment, the movement caused the iron roughneck roll to slip off and fall through the vent opening of the Traction Motor A.

The roll dropped 10 metres to the area below into a barricaded area. No personnel were present in the barricaded area and no injuries or damages resulted from the dropped object.

Dropped object weight = 3.4kg.

WHAT WENT WRONG:

Aligning Traction Motor A with the input shaft A on the driller side gear box:

- Area under the substructure barricaded off to avoid passing people underneath the sub-base.
- Traction motor coupling cover removed but the hole on the floor left uncovered.

Using jack to align to push the traction motor:

- The ¼ part of Iron roughneck roll slipped.
- A piece of wood and the ¼ part of Iron roughneck roll were being used instead of using one piece of proper support.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

- Support was badly installed with two pieces of material used to support instead of one. Friction of the Iron Roughneck roll and the jack caused the dropped object to slip.
- Preparation/Organization, the area below the substructure was barricaded off to prevent persons passing through but the hole on the drill floor was left uncovered.
- The JSA was not followed, work area not checked and the hole on the drill floor was left uncovered.
- Stop work authority not used by involved people.

CAUSAL FACTORS:

PEOPLE (ACTS): Inattention/Lack of Awareness: Improper decision making or lack of judgment

PROCESS (CONDITIONS): Organizational: Inadequate work standards/procedures

PROCESS (CONDITIONS): Organizational: Inadequate hazard identification or risk assessment

MIDDLE EAST ONSHORE

DATE: 01 Dec 2022

COUNTRY: Kurdistan Region of Iraq

FUNCTION: Drilling

CAUSE: Dropped objects

ACTIVITY: Drilling, workover, well operations

RULE: Working at height

NARRATIVE:

Coiled tubing operators were in the process of rigging up a Coiled Tubing (CT) Injector Head on a rig's travelling block using a crane. One operator was inside a man basket holding the slings and the other one was standing on the injector head attaching the shackle. After successfully attaching one sling, the CT operators attempted to attach the second sling. While pulling out the shackle pin, it fell approximately 9 m. to the rig floor. The pin was 20 cm long and weighed 5.9 kg.

WHAT WENT WRONG:

- Shackle size made it difficult to fit two slings which led to repeated attempts and extended handling of the removable shackle pin.
- The man basket and crane combination did not offer ample manoeuvring space for easy access.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

- Perform risk assessment and hazard identification for working at height before conducting the task.
- Clear/barricade the area beneath the activity (no one permitted in the drop zone area).
- Static tethered tool kits to be made available for any task while working at height.
- Confirm the suitability of man riding basket, lifting auxiliaries, and crane capability before performing the task.
- Work should be stopped for reassessment whenever conditions change.

CAUSAL FACTORS:

PROCESS (CONDITIONS): Organizational: Inadequate work standards/procedures

PROCESS (CONDITIONS): Organizational: Inadequate hazard identification or risk assessment

PROCESS (CONDITIONS): Organizational: Inadequate supervision

DATE: 21 May 2022

COUNTRY: Kurdistan Region of Iraq

FUNCTION: Drilling

CAUSE: Dropped objects

ACTIVITY: Drilling, workover, well operations

RULE: Bypassing safety controls

NARRATIVE:

Top drive system (TDS) link tilt hydraulic cylinders pins and retainer was found near the V-door. It weighed 360 grams and fell from a height of between 10 and 38 metres, as the height it fell from could not be determined precisely.

WHAT WENT WRONG:

Contractor was aware of the Rig TDS manufacturer safety alerts regarding the problem with the link tilt system pins, but they did not take necessary action and the TDS link tilt was not included in their Preventative Maintenance System (PMS) inspection regime.

MIDDLE EAST ONSHORE

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

- Contractor to include Rig TDS link tilt cylinder pin inspections in their PMS inspection regime.
- There was design change to mitigate this issue. A newer style of retaining pin and retaining nut had been introduced to mitigate this concern.

CAUSAL FACTORS:

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate/defective tools/equipment/materials/products

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate maintenance/inspection/testing

PROCESS (CONDITIONS): Organizational: Failure to report/learn from events

DATE: 18 May 2022

COUNTRY: Kurdistan Region of Iraq

FUNCTION: Drilling

CAUSE: Pressure release

ACTIVITY: Drilling, workover, well operations

RULE: Bypassing safety controls

NARRATIVE:

During rigging down of well testing equipment, an atmospheric storage tank was over pressured using compressed air from a compressor via a manifold. This over pressure blew the tank roof off the body of the tank, which landed in a nearby wastewater pit.

WHAT WENT WRONG:

- Supervisor purged the tank with air which was non-compliance to the standard operating procedure.
- 2" vent lines on the tanks were welded shut without MoC documentation.
- HAZOP did not recognize over pressure as an event in the atmospheric tanks.

Process Safety Fundamental: We stay within operating limits. We apply procedures. We respect hazards.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

- Develop a procedure for contracting and the use of atmospheric and low-pressure tanks/Vessels.
- All 3rd Party contractors personnel training and competency checks to be done before start of operations and ensure they are competent.

CAUSAL FACTORS:

PEOPLE (ACTS): Use of Tools, Equipment, Materials and Products: Improper use/position of tools/equipment/materials/products

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate design/specification/management of change

PROCESS (CONDITIONS): Organizational: Inadequate training/competence

PROCESS (CONDITIONS): Organizational: Inadequate work standards/procedures

PROCESS (CONDITIONS): Organizational: Inadequate hazard identification or risk assessment

MIDDLE EAST ONSHORE

DATE: 05 Mar 2022

COUNTRY: Kurdistan Region of Iraq

FUNCTION: Drilling

CAUSE: Struck by (not dropped object)

ACTIVITY: Drilling, workover, well operations

RULE: Line of fire

NARRATIVE:

While the floorman was using chain tongs to remove a sub from the drill string, the driller activated the top drive system (TDS) causing the drill string to turn while the floorman was holding the chain tongs. The Floorman was thrown onto the rig floor when the Driller operated TDS.

WHAT WENT WRONG:

The Driller should not have engaged the TDS when there were chain tongs attached. No one, including the four senior supervisory personnel, intervened with the STOP work authority when the Floorman stood in the line of fire on the rotating chain tong.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

Floor crew and driller to carry out toolbox talks, highlighting line of fire and correct tripping techniques, prior to commencing any tripping operations according to the Risk Assessment/Standard Operating Procedure.

Always reinforce the Stop Work Authority.

CAUSAL FACTORS:

PEOPLE (ACTS): Following Procedures: Improper position (in the line of fire)

PEOPLE (ACTS): Use of Protective Methods: Failure to warn of hazard

PROCESS (CONDITIONS): Organizational: Inadequate training/competence

PROCESS (CONDITIONS): Organizational: Inadequate hazard identification or risk assessment

PROCESS (CONDITIONS): Organizational: Inadequate supervision

DATE: 10 Jan 2022

COUNTRY: Kurdistan Region of Iraq

FUNCTION: Drilling

CAUSE: Struck by (not dropped object)

ACTIVITY: Lifting, crane, rigging, deck operations

RULE: Safe mechanical lifting

NARRATIVE:

While the air winch hook was clipped onto the lifting eye bolt on the body of the air winch, the winch then continued to pull up and the air winch pulled itself off the base pedestal welded connection.

WHAT WENT WRONG:

- Winch control lever was not functioning correctly.
- Deficiencies had been identified on this winch during inspection in the post move inspection.
- No Standard operating procedure in place.
- No formal training or records kept for Winch Operators.

MIDDLE EAST ONSHORE

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

- Ensure Standard Operating Procedures are in place for this type of activity.
- Ensure formal training is provided for air winch operators.

CAUSAL FACTORS:

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate maintenance/inspection/testing

PROCESS (CONDITIONS): Organizational: Inadequate training/competence

PROCESS (CONDITIONS): Organizational: Inadequate work standards/procedures

PROCESS (CONDITIONS): Organizational: Inadequate hazard identification or risk assessment

PROCESS (CONDITIONS): Organizational: Inadequate supervision

DATE: 01 Mar 2022

COUNTRY: Kurdistan Region of Iraq

FUNCTION: Drilling

CAUSE: Struck by (not dropped object)

ACTIVITY: Lifting, crane, rigging, deck operations

RULE: Bypassing safety controls

NARRATIVE:

Operator initiated the wrong control lever and the auxiliary hook did not stop, which led to the auxiliary hook striking the turn down sheave, flipping over and landing on the back of the boom. The turn down sheave and associated components were damaged.

WHAT WENT WRONG:

- Crane operator operated the wrong control panel and was distracted and not focused on the hook block assembly.
- Crane auxiliary line was not included in the 3rd party inspection.
- 3rd Party inspector was not suitably qualified to carry out the inspection.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

- All 3rd Party contractors personnel training and competency checks to be done before start of operations.
- Always reinforce the Stop Work Authority.

CAUSAL FACTORS:

PEOPLE (ACTS): Inattention/Lack of Awareness: Lack of attention/distracted by other concerns/stress

PROCESS (CONDITIONS): Protective Systems: Inadequate/defective warning systems/safety devices

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate/defective tools/equipment/materials/products

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate maintenance/inspection/testing

PROCESS (CONDITIONS): Organizational: Inadequate training/competence

MIDDLE EAST ONSHORE

DATE: 08 Nov 2022

COUNTRY: Kurdistan Region of Iraq

FUNCTION: Production

CAUSE: Struck by (not dropped object)

ACTIVITY: Transport - Land

RULE: Work authorization

NARRATIVE:

An 80 ton crane and operator had completed a lifting operation at the production facility for the installation of a new security cabin, and were due to return to the town, when the driver received a call from the crane owner requiring him to relocate both himself and the crane to a well inside our operation block, due to a hose leak identified on an existing crane at the rig site (another crane used at rig site). On route, the crane's brakes started to smoke due to the steep decline from the well site road turning to the base of the mountain, where the driver stopped to let the brakes cool down for approximately 33 minutes. The driver then restarted the crane and continued his journey.

The route was a total distance of 31.2 kilometres. The first stop was after 3.1 kilometres calculated from the wellsite turning to the base of the mountain due to the crane's brakes starting to smoke. The driver continued his journey a further 26.1 kilometres over inclining and declining roads before approaching the turning. As the driver reached the turning, he found that the brakes would not function and that the crane was moving too fast to make the right turn. A split-second decision was made to continue down the declining road hoping that the road would level out. The crane driver was unaware that the well site was located 2 kilometres ahead at the bottom of the hill. On approach to the well site, the crane crashed through both the security barrier and perimeter gates, continued across the pad, colliding with two concrete barriers used to protect the flowline, before finally coming to a stop above the flowline after pushing the line off its supporting brackets.

WHAT WENT WRONG:

Immediate Cause:

Brake failure due to an incorrectly set up braking system reducing the crane's braking power to less than 40%.

Basic Cause:

The sub-standard setup of the crane's braking system on August 6 and 7 meant that only three of its eight wheels were fully functional, severely reducing its stopping power.

The crane driver did not hold a driving license, heavy plant license or crane training certificate.

Root Cause:

No contractor assurance systems in place pre/post contract award. A system for auditing and inspecting contractors and sub-contractors would assure that compliance with contractual obligations is being fulfilled before contract award and throughout the contract's lifetime.

Process Safety Fundamental: We stop if the unexpected occurs.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

- Implementing third-party audits and inspections of contractors and sub-contractors confirms contractual and legal compliance in line with a suitable standard, such as ISO 9001. This program would need to include pre-contract and post-contract awards.

MIDDLE EAST ONSHORE

- Enhance operational controls to include inspections of driver's licenses with specific attention to driving categories.
- Assessment of security checkpoints confirming suitability of controls.
- Company to consider the contract hire of a full-time 80Ton crane and two crane operators on a rotational basis. Hiring a crane and operator would significantly reduce the need to hire cranes externally while ensuring compliance with Local and National Laws.

CAUSAL FACTORS:

PEOPLE (ACTS): Inattention/Lack of Awareness: Improper decision making or lack of judgment

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate maintenance/inspection/testing

DATE: 25 Feb 2022

COUNTRY: Kurdistan Region of Iraq

FUNCTION: Construction

CAUSE: Dropped objects

ACTIVITY: Construction, commissioning, decommissioning

RULE: Bypassing safety controls

NARRATIVE:

The secondary roof (25m x 9.5m) of the newly constructed laundry building became detached due to extreme high winds, traveling approximately 50m, before landing on the main camp generators, resulting in a minor diesel spill (diesel leak contained within bund and therefore not categorized as a spill).

WHAT WENT WRONG:

The Roofing Subcontractor:

- failed to follow manufacturer's drawings and procedures.
- failed to use 5x38 fixings to anchor the canopy to the laundry building chassis.
- used tack welds as an alternative/unapproved method of joining the structures together.

High wind speeds of up to 78.6km created the energy to lift the canopy structure from the building.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

- Contractor Selection and Oversight - Contractor pre-qualification process not effective: a pre-qualification process was in place and utilized, but it did not identify relevant deficiencies in the Contractor's capabilities.
- Subcontractor Management - Lack of control by the Principal Contractor, who mobilized a roofing subcontractor without performing competence verification or due diligence checks.
- Inadequate work oversight (company/contractor) - Company had limited resources to cover the Camp Construction activities. Main contractor failed to provide proper oversight of the Subcontractor's work activities.

CAUSAL FACTORS:

PEOPLE (ACTS): Use of Protective Methods: Equipment or materials not secured

PROCESS (CONDITIONS): Organizational: Inadequate work standards/procedures

PROCESS (CONDITIONS): Organizational: Inadequate supervision

MIDDLE EAST ONSHORE

DATE: 14 Nov 2022

COUNTRY: Oman

FUNCTION: Drilling

CAUSE: Dropped objects

ACTIVITY: Lifting, crane, rigging, deck operations

RULE: Safe mechanical lifting

NARRATIVE:

While lowering the Rig mast, the mast retention slings snapped, causing the traveling block to move from the retention position and hang by the drill line itself in a lower position. High potential of drop object.

WHAT WENT WRONG:

- Improper active monitoring while lowering the mast and failed to capture the block while getting overpulled.
- The best practice of securing the block is not identified due to lack of procedural original equipment manufacture information.
- Lack of intervention (SWA) and reporting system by crew when there was doubt about the correct block securing method.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

- Procedures and JSAs relevant to lowering & raising mast including parking and securing travelling block to be reviewed and updated.
- Roll out and train the crews on the new block securing method after it has been agreed on and finalized to ensure new knowledge and skills are acknowledged & transferred effectively.
- Dedicate specific set of slings to be used uniquely for the block retention and to be secured in a separate box under toolpusher custody to avoid mixing with other rigging loft or choosing wrong type of retention slings.

CAUSAL FACTORS:

PEOPLE (ACTS): Use of Tools, Equipment, Materials and Products: Improper use/position of tools/equipment/materials/products

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate design/specification/management of change

PROCESS (CONDITIONS): Organizational: Inadequate hazard identification or risk assessment

DATE: 29 May 2022

COUNTRY: Oman

FUNCTION: Production

CAUSE: Caught in, under or between (excl. dropped objects)

ACTIVITY: Transport - Land

RULE: Driving

NARRATIVE:

A fully loaded brackish tanker slipped backward and got stuck on the edge of the hill while driving to a well location with a high potential chance of rolling over. No injuries reported.

MIDDLE EAST ONSHORE

WHAT WENT WRONG:

- The trailer tanker brakes were not working properly (Break Hydraulic pressure/ brakes boosters have air pressure leaks).
- The slope height does not help stop the trailer tanker in case of a slippery situation.
- The tyres were covered with mud, making holding onto the hill slope road almost impossible.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

- Increase the awareness of stop work authority among the drivers in case facing unsafe road condition.
- The driver must inspect the vehicle daily before starting the work and inform the supervisor about the vehicle issues.
- The pits need proper supervision to check the condition of the roads/entrance.

CAUSAL FACTORS:

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate design/specification/management of change

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate maintenance/inspection/testing

PROCESS (CONDITIONS): Work Place Hazards: Inadequate surfaces, floors, walkways or roads

DATE: 02 Oct 2022

COUNTRY: Oman

FUNCTION: Construction

CAUSE: Struck by (not dropped object)

ACTIVITY: Construction, commissioning, decommissioning

RULE: Bypassing safety controls

NARRATIVE:

A High-pressure Rental pump has been mobilised to injection well in an attempt to increase the water injection rate using a High Pressure Pump system. During pre-commissioning, a water hose developed a Leak due to the impingent of a high velocity water jet.

WHAT WENT WRONG:

- The supervisor did not properly control the velocity of the fluid while opening the choke valve.
- Inadequate protective device: pressure gauge was not in place to monitor and control the pressure and high velocity of water jetting.
- Improper communication between company team and contractor team during PTW process. PTW did not mention that the valve was open, however, opening gate valves was communicated by phone.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

- Flexible hoses to be replaced with Steel pipe and any future HP Pump units.
- Drilling Contractors/Service companies require close company supervision to ensure compliance with company standards and procedures.
- Permit to Work and LOTO awareness training to be provided to new contractors to the Field.
- Formalise communication between company and contractor– written Procedures should be developed for operational phase and communication numbers to be shared.

MIDDLE EAST ONSHORE

CAUSAL FACTORS:

PEOPLE (ACTS): Use of Tools, Equipment, Materials and Products: Servicing of energized equipment/inadequate energy isolation

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate design/specification/management of change

PROCESS (CONDITIONS): Organizational: Inadequate hazard identification or risk assessment

PROCESS (CONDITIONS): Organizational: Inadequate communication

PROCESS (CONDITIONS): Organizational: Inadequate supervision

DATE: 11 Jul 2022

COUNTRY: UAE

FUNCTION: Construction

CAUSE: Caught in, under or between (excl. dropped objects)

ACTIVITY: Lifting, crane, rigging, deck operations

RULE: Safe mechanical lifting

NARRATIVE:

Personal injury (laceration) in the right ear while loading plates into the load basket.

WHAT WENT WRONG:

The PTW system was not followed.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

Provide supervision and high-level oversight during all work activities. Enforcement of the control of work system.

CAUSAL FACTORS:

PEOPLE (ACTS): Inattention/Lack of Awareness: Improper decision making or lack of judgment

MIDDLE EAST OFFSHORE

DATE: 20 Jan 2022

COUNTRY: Qatar

FUNCTION: Drilling

CAUSE: Falls from height

ACTIVITY: Maintenance, inspection, testing

RULE: Working at height

NARRATIVE:

The IP was working on 2nd level of port aft leg jacking unit to conduct maintenance activities. His supervisor was preparing tools on deck, and while the IP was about to lift the jerry can of oil over the handrail (approximately 19kg), suddenly the handrail failed, and the IP fell to the deck from a height of approximately 5.5 m. The medic was immediately notified, a stretcher team arrived and transferred the IP to clinic. The IP was conscious but in shock. Urgent MEDEVAC onshore arranged.

WHAT WENT WRONG:

- Heavily corroded handrail was not identified.
- No Preventive Maintenance System (PMS) in place for inspection/testing of handrail.
- IP leaning on handrail for lifting jerry-can was not part of the job safety analysis and this task not identified as Work at Height.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

- Launch handrail inspection.
- Ensure all surrounding handrails are checked and discussed during Toolbox talk prior starting new job. Concerned JSA to be updated.
- Maintain handrails register up to date with established preventative maintenance system.
- Implement annual third-party handrail inspection and testing.

CAUSAL FACTORS:

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate maintenance/inspection/testing

PROCESS (CONDITIONS): Organizational: Inadequate hazard identification or risk assessment

DATE: 23 Oct 2022

COUNTRY: Qatar

FUNCTION: Production

CAUSE: Dropped objects

ACTIVITY: Lifting, crane, rigging, deck operations

RULE: Safe mechanical lifting

NARRATIVE:

Five riggers were assigned to move the gratings one by one from the bundle which was placed vertically on a pallet located nearby the handrail on the walkway of middle deck. The metal strap that held the grating compressed against each other was removed and the grating began to slide on its side. A piece of grating slid from the bundle through the handrails and fell from 19 metre height on the HVAC unit (2 metre height) located at the main deck bounced and fell on the walkway. Two workers were located approximately 5m away from the HVAC Unit. A low-pressure fuel gas line was located approximately 2m away of the impacted area.

MIDDLE EAST OFFSHORE

WHAT WENT WRONG:

Dropped Object occurred due to improper task risk assessment and execution of activities.

Fortunately, the grating fell on to a HVAC unit, thus absorbing the energy before coming to rest on the walkway. Area below was not barriered off as it was not considered as being in the drop zone during the initial lifting operation.

Process Safety Fundamental: We respect hazards

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

Grating of various size to be packed to ensure small sections are placed on outside of bundle. Such lifts shall be considered as routine lifts with appropriate lift plan. When a task changes, stop the job and re-assess additional risk.

CAUSAL FACTORS:

PEOPLE (ACTS): Following Procedures: Overexertion or improper position/posture for task

PEOPLE (ACTS): Following Procedures: Improper lifting or loading

PROCESS (CONDITIONS): Organizational: Inadequate hazard identification or risk assessment

DATE: 04 Oct 2022

COUNTRY: Qatar

FUNCTION: Construction

CAUSE: Falls from height

ACTIVITY: Construction, commissioning, decommissioning

RULE: Working at height

NARRATIVE:

The IP was working inside the welding habitat at 1.8m high supported by a scaffolding platform. He was standing on a narrow platform beside an open hole. There were two fire blankets on top of each other near the open hole. IP inadvertently stepped on the fire blanket, slipped and fell through the open hole to the lower deck of the habitat.

WHAT WENT WRONG:

- Failure to provide proper assessment and work method.
- Stop Work Authority and Duty to Step In were not implemented.
- Lack of safety awareness.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

- Generate a method statement that should be issued prior preparing a Permit to Work.
- Clarify stages and requirements of all tasks and include potential changes.
- Any amendments to scaffold during task must be reassessed and authorized by authorized personnel.
- Use of Fall Arrest Equipment whenever working near an open edge.

CAUSAL FACTORS:

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate design/specification/management of change

PROCESS (CONDITIONS): Organizational: Inadequate hazard identification or risk assessment

NORTH AMERICA ONSHORE

DATE: 09 Sep 2022

COUNTRY: Canada

FUNCTION: Drilling

CAUSE: Dropped objects

ACTIVITY: Lifting, crane, rigging, deck operations

RULE: Line of fire

NARRATIVE:

While lifting a rotator head, the tugger line snagged on a beam and snapped. The rotator head fell approximately 10ft to the mat, bounced and landed approximately 2ft from a worker. The tugger line also recoiled, went through the crown and fell approximately 120ft to the floor.

WHAT WENT WRONG:

A problem with the tugger line was not anticipated, and when it occurred, it snagged on the I-Beam. Hazards associated with this activity were not anticipated and reactive controls were not implemented.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

Expanding hazard assessments to include areas and potential sources of dropped objects, including assessment of drop zones, associated with rig activities.

CAUSAL FACTORS:

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate design/specification/management of change

PROCESS (CONDITIONS): Organizational: Inadequate work standards/procedures

DATE: 06 Mar 2022

COUNTRY: Canada

FUNCTION: Drilling

CAUSE: Exposure electrical

ACTIVITY: Drilling, workover, well operations

RULE: Energy isolation

NARRATIVE:

During rig out of the service rig, a worker was attempting to coil and stow a temporary electrical cable. The cable end became stuck underneath the catwalk. A worker attempted to pull the cable end free but was unsuccessful. The worker then pushed the cable end in an attempt to free the cable end from the catwalk. The workers right hand (with donned wet glove) made contact with the inner connections (4 pole/208 Volts/60 Amps). The worker reported experiencing a momentary electrical shock. The worker reported pain continuing post shock with tingling sensations.

The ground conditions were snow covered with water accumulation in low spots atop frozen ground with temperatures of +5 Degrees Celsius with clear skies.

WHAT WENT WRONG:

Contractor failed to demonstrate a systematic process for identifying and implementing controls for all electrical hazards in their operations. Result: engineering controls were non-existent; and weak admin controls.

NORTH AMERICA ONSHORE

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

Engineering Controls

- All electrical panels on service rigs must have the “Plug End” connection. The power cord must have the “Socket End” part of the connection.
- There needs to be a physical lock and LOTO procedure for power cord at the rig electrical panel.

Admin Controls:

There is a need to create a Work Instruction for energy isolation, installation/removal and securement of the service rig powered systems.

Contractor HSSE Management:

Expectations shall be set for Contractors to provide their audit plan/schedule, and to review results/insights at appropriate frequency.

CAUSAL FACTORS:

PEOPLE (ACTS): Use of Tools, Equipment, Materials and Products: Servicing of energized equipment/inadequate energy isolation

PROCESS (CONDITIONS): Organizational: Inadequate work standards/procedures

PROCESS (CONDITIONS): Organizational: Inadequate hazard identification or risk assessment

DATE: 12 Mar 2022

COUNTRY: Canada

FUNCTION: Drilling

CAUSE: Struck by (not dropped object)

ACTIVITY: Transport - Land

RULE: Driving

NARRATIVE:

Two vehicles, a pickup truck and empty tractor trailer, collided on a winter ice road curved corner. Major damage was sustained to the pickup and minor damage to the tractor trailer unit.

WHAT WENT WRONG:

The driver of the pickup was over driving road conditions and unable to maintain vehicle control or stop in time to avoid collision.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

An opportunity to incorporate risk assessment into annual IWP planning to prevent the creation of blind corners.

CAUSAL FACTORS:

PEOPLE (ACTS): Following Procedures: Work or motion at improper speed

NORTH AMERICA ONSHORE

DATE: 05 Dec 2022

COUNTRY: Canada

FUNCTION: Construction

CAUSE: Dropped objects

ACTIVITY: Lifting, crane, rigging, deck operations

RULE: Safe mechanical lifting

NARRATIVE:

A crane was extending their boom out to 70ft to prepare for a personnel lift. When the boom reached approximately 60ft the whip line ball crowned causing the approximately 125lb ball to separate from the cable and drop to the trailer deck, landing approximately 8ft from a worker.

WHAT WENT WRONG:

The master bypass for the anti-blocking system was left engaged after being used by a previous operator while rigging out. Pre-use inspections were not completed, and it was common to ignore alarms. Exclusion zones had not yet been established.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

Ensure work planning and hazard assessments address potential sources of dropped objects. Ensuring effective safety bypass processes are in place, while removing the ability to engage bypasses without appropriate authorization.

CAUSAL FACTORS:

PEOPLE (ACTS): Use of Tools, Equipment, Materials and Products: Improper use/position of tools/equipment/materials/products

PEOPLE (ACTS): Use of Protective Methods: Inadequate use of safety systems

PROCESS (CONDITIONS): Organizational: Inadequate work standards/procedures

DATE: 13 Jan 2022

COUNTRY: USA

FUNCTION: Drilling

CAUSE: Caught in, under or between (excl. dropped objects)

ACTIVITY: Drilling, workover, well operations

RULE: Line of fire

NARRATIVE:

The rig crew and casing crew were in the process of making up a joint of casing when they noticed the backup tongs were on the casing collar and needed to be lowered to the casing joint body. The IP decided to help push the backup tongs down by standing on them. He placed his right hand on the stationary equipment and his left hand on the casing tongs to hold himself steady and push down on the backup tongs. The IP had inadvertently put his left hand on the casing tong door latch and bent his finger down just as the tong operator had started making up the joint of casing. As the casing tong began to turn, the IP's finger was caught between the gear and tong body, severing his left ring finger at the third knuckle (closest to fingernail).

NORTH AMERICA ONSHORE

WHAT WENT WRONG:

The IP had inadvertently put his left hand on the casing tong door latch and bent his finger down just as the tong operator had started making up the joint of casing. As the casing tong began to turn, the IP's finger was caught between the gear and tong body, severing his left, ring finger at the third knuckle.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

n/a

CAUSAL FACTORS:

PEOPLE (ACTS): Following Procedures: Improper position (in the line of fire)

PEOPLE (ACTS): Following Procedures: Improper lifting or loading

PEOPLE (ACTS): Inattention/Lack of Awareness: Improper decision making or lack of judgment

DATE: 10 Dec 2022

COUNTRY: USA

FUNCTION: Drilling

CAUSE: Dropped objects

ACTIVITY: Drilling, workover, well operations

RULE: Line of fire

NARRATIVE:

Rig was tripping in the hole, Top Drive raised in the blocks, preparing to make a connection. Floorhand #2 saw Floorhand #1 (Injured Person) struck by a bright object on his upper left arm (Biceps), while standing in the alley / rig floor with a stand of drill pipe in his tailrope. Fallen object is a steel Drag Chain link rod measuring approximately 3/8" wide x 16" long, weighing 396-g (0.873-lbs), and was positively identified as having fallen from the Drag Chain belt from a minimum height of 27.43-m (90-ft) above the rig floor. Injured Person was alert, responsive, and transported to hospital for further evaluation.

WHAT WENT WRONG:

- Drag Chain wear and tear.
- Secondary Retention to link rod failed.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

- Review and follow proper DROPS inspection requirements and frequency.
- Ensure personnel performing DROPS inspection are properly trained to recognize DROPS hazards.
- Add verification criteria for Well Site Supervisor to all completed DROPS inspections.
- Evaluate Drag Chain design and installation, as well as transitioning rig to a dedicated service loop.

CAUSAL FACTORS:

PEOPLE (ACTS): Use of Protective Methods: Equipment or materials not secured

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate/defective tools/equipment/materials/products

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate maintenance/inspection/testing

NORTH AMERICA ONSHORE

DATE: 08 Oct 2022

COUNTRY: USA

FUNCTION: Drilling

CAUSE: Dropped objects

ACTIVITY: Lifting, crane, rigging, deck operations

RULE: Safe mechanical lifting

NARRATIVE:

A rig contractor was rigging down from a well, and the logistics contractor was in the process of removing the V-door from the substructure. The lifting equipment was secured, with a tagline attached. Pins securing the V door to the substructure were removed to allow lifting and removal of the V-door. As the load was being lifted, the crane reversed, and the load levelled. This created a bind in the load, causing one of the d-rings to dislodge from the lifting hook. The V-door fell approximately 3 metres onto the drill line and managed pressure drilling line below.

All personnel were clear of the area at the time of the event, and no injuries were reported.

WHAT WENT WRONG:

Inadequate supervision:

- Supervisory personnel did not ensure crane inspection was conducted prior to the mechanical lift.
- Supervisory personnel did not ensure rigging equipment was adequate for the task.
- Onsite supervision failed to identify and properly manage onsite operational hazards and risks.

Deviation from Procedures by Crew:

- Onsite personnel did not follow proper lifting best practices.

Inadequate Hazard Identification:

- Onsite personnel failed to identify and properly manage mechanical lifting risks.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

- Survey of Reiteration of Policy/Procedure Utilization: Management of the contractor reiterated to personnel the existing procedures for Crane lifting operations and confirm personnel understand how to apply these procedures.
- Equipment Replacement: Contractor personnel replaced the latch on the lifting hook and implemented additional (4 leg chain sling) to avoid improper lifting angles caused by load levelling.
- Post Incident Toolbox Talk and Sharing Lessons Learned: Shortly following the incident, onsite meeting was held with teams from both company and contractors. Event was discussed in detail to fully understand causes, inadequacies, and where utilization of procedures and safe work practices need to be applied to prevent reoccurrence. Expectation of following established mechanical lifting procedures was re-emphasized.
- Crane Inspection: Daily crane inspection conducted and documented prior to resuming operations.
- Retraining Recommendation: Company requested contractor to conduct refresher training to its personnel in identifying, assessing, and mitigating hazards/risks as they apply to specific operations.

CAUSAL FACTORS:

PEOPLE (ACTS): Following Procedures: Deviation unintentional (by individual or group)

NORTH AMERICA ONSHORE

DATE: 04 Oct 2022

COUNTRY: USA

FUNCTION: Drilling

CAUSE: Dropped objects

ACTIVITY: Lifting, crane, rigging, deck operations

RULE: Bypassing safety controls

NARRATIVE:

The boom pole hoist was secured to its tie off point on the casing porch with its handle inadvertently stuck in the hoist position. The hydraulic power unit was engaged, and the boom pole spool began to hoist. The casing porch was lifted from its cantilever support arms until the boom pole hoist cable parted and ended up suspended on one of the arms.

WHAT WENT WRONG:

- Lock out Tag Out Not utilized.
- No breakaway or controlled mechanism was in place to prevent lifting of porch.
- Hoist lever malfunctioning.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

Relocate anchor point from casing porch to right floor, utilize breakaway straps to affix hoist to anchor point. Update rig move checklist to include independent verification of porch bolt installation.

CAUSAL FACTORS:

PEOPLE (ACTS): Use of Tools, Equipment, Materials and Products: Servicing of energized equipment/inadequate energy isolation

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate design/specification/management of change

DATE: 22 Feb 2022

COUNTRY: USA

FUNCTION: Drilling

CAUSE: Dropped objects

ACTIVITY: Lifting, crane, rigging, deck operations

RULE: Safe mechanical lifting

NARRATIVE:

Brakes failed to hold the load allowing an uncontrolled decent of the blocks and tubing. As the load dropped the blocks came to rest against the handrail of the work platform near one of the rig employees.

WHAT WENT WRONG:

- Brake linkage was adjusted by a rig crew that were not knowledgeable or trained to complete the task, resulting in brakes not performing as designed.
- Lack of communication concerning unusual rig adjustments from crew to onsite leadership.

NORTH AMERICA ONSHORE

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

- Retrain on brake adjustments and recognition of brake issues with crew and person in charge.
- Ensure quality closeout meetings held at end of each shift enhancing communication of any known issues.
- Establish protocol that no adjustments are made unless operator and pusher are present.
- Reinforce Stop Work Authority.

CAUSAL FACTORS:

PEOPLE (ACTS): Following Procedures: Deviation unintentional (by individual or group)

PEOPLE (ACTS): Use of Protective Methods: Disabled or removed guards, warning systems or safety devices

PEOPLE (ACTS): Inattention/Lack of Awareness: Improper decision making or lack of judgment

PEOPLE (ACTS): Inattention/Lack of Awareness: Lack of attention/distracted by other concerns/stress

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate design/specification/management of change

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate maintenance/inspection/testing

PROCESS (CONDITIONS): Work Place Hazards: Congestion, clutter or restricted motion

PROCESS (CONDITIONS): Organizational: Inadequate training/competence

PROCESS (CONDITIONS): Organizational: Inadequate work standards/procedures

PROCESS (CONDITIONS): Organizational: Inadequate hazard identification or risk assessment

PROCESS (CONDITIONS): Organizational: Inadequate communication

PROCESS (CONDITIONS): Organizational: Inadequate supervision

DATE: 05 Jul 2022

COUNTRY: USA

FUNCTION: Drilling

CAUSE: Exposure noise, chemical, biological, vibration, extreme temperature

ACTIVITY: Drilling, workover, well operations

RULE: Work authorization

NARRATIVE:

After night shift completed the blender swap, the 1 inch chemical hose for the biocide was left unsecured inside secondary containment by the blender.

While transferring biocide chemical, the chemical hose was turned on releasing an estimated 2 barrels of the biocide chemical inside secondary containment and 14 barrels outside of containment onto the ground.

WHAT WENT WRONG:

The spill was not detected until after more than 6 hours. Personnel not familiar / not trained regarding response to spills or releases to the environment.

Both chem add and operator did not perform the communication and verification of the flow.

Replacing equipment between crew change, repetitive technical issues, new supervisor unfamiliar with staff experience.

Process Safety Fundamental: We Apply Procedures.

NORTH AMERICA ONSHORE

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

- Reinforce incident reporting and stop work authority.
- Assess current competency profile for personnel.
- Provide training based on the competency assessment for everyone.
- Amend/update existing procedure to include verification process when changing/swapping blenders throughout check list.
- Develop check list for replacing/swapping blender.
- Establish a hand over process (document) to establish proper handover between changing crews.

CAUSAL FACTORS:

PEOPLE (ACTS): Inattention/Lack of Awareness: Lack of attention/distracted by other concerns/stress

PROCESS (CONDITIONS): Organizational: Inadequate training/competence

PROCESS (CONDITIONS): Organizational: Inadequate work standards/procedures

PROCESS (CONDITIONS): Organizational: Inadequate hazard identification or risk assessment

DATE: 15 Aug 2022

COUNTRY: USA

FUNCTION: Drilling

CAUSE: Struck by (not dropped object)

ACTIVITY: Drilling, workover, well operations

RULE: Line of fire

NARRATIVE:

The rig crew was tripping out of the hole. The rig employee (IE) was pushing an 8" drill collar away from the stump to position it into the racking board. The drill collar became hung up on the derrick board.

The IE repositioned and attempted to pull the drill collar into the racking board. As the IE was moving the drill collar, the driller slacked off the winch connected to the collar to help move it around the derrick board. As the drill collar was coming down, the IE's foot slipped on the rig floor. This slip caused his right foot to slide underneath the drill collar as it descended, causing a major injury to the toe.

WHAT WENT WRONG:

The floor-hand was pulling the drill collar instead of pushing the drill collar. No effective means of communication established for blind lifts.

Slip hazard presented by the rig floor perceived to be a low risk. No control identified beyond cleaning the floor periodically.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

- Review and update JSA with specific learning from this incident in relation to slips, trips and falls and ensure appropriate controls are in place.
- Communicate updated JSA. Ensure all relevant parties understand associated hazards and controls prior commencing the activities.

NORTH AMERICA ONSHORE

CAUSAL FACTORS:

PEOPLE (ACTS): Following Procedures: Improper position (in the line of fire)

PEOPLE (ACTS): Use of Protective Methods: Failure to warn of hazard

PROCESS (CONDITIONS): Organizational: Inadequate hazard identification or risk assessment

DATE: 12 Jan 2022

COUNTRY: USA

FUNCTION: Production

CAUSE: Explosion, fire or burns

ACTIVITY: Maintenance, inspection, testing

RULE: Line of fire

NARRATIVE:

A flash fire occurred when a contractor, the Injured Person (IP), was blowing down the surface casing pressure. Proper notifications were made, and the location was shut in. The events were as follows:

- Vent off casing gas as part of routine rounds.
- Opened manual valve and flash fire occurred.
- Shut valve and called for help.
- Lead Operator attended to IP.
- IP received medical attention.

WHAT WENT WRONG:

Standard Operating Procedure (SOP) for Casing Pressure Reduction was not complete nor implemented through training. Management of Change (MOC) could not be located for casing pressure equipment change. Insufficient risk assessment and hazard mitigation. Inadequate collaboration and unclear roles/responsibilities. Safety concerns not raised and/or not addressed. Inadequate technical barriers to prevent pressure build-up.

Process Safety Fundamentals: We apply procedures. We recognise change. We respect hazards.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

Key learning for other units:

- SOP out of date, not used.
- MOC process not robust enough.
- Risk assessments needed for high-risk work.
- Leadership engagement needed.

CAUSAL FACTORS:

PEOPLE (ACTS): Use of Tools, Equipment, Materials and Products: Improper use/position of tools/equipment/materials/products

PROCESS (CONDITIONS): Protective Systems: Inadequate/defective guards or protective barriers

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate design/specification/management of change

PROCESS (CONDITIONS): Organizational: Inadequate work standards/procedures

PROCESS (CONDITIONS): Organizational: Inadequate hazard identification or risk assessment

NORTH AMERICA ONSHORE

DATE: 18 Oct 2022

COUNTRY: USA

FUNCTION: Production

CAUSE: Explosion, fire or burns

ACTIVITY: Production operations

RULE: Other

NARRATIVE:

Battery fire.

WHAT WENT WRONG:

Emergency shutdown (ESD) on wells failed to close and water tanks ran over sending fluid to combustor starting initial fire.

Process Safety Fundamental: We sustain barriers.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

Retrofit existing ESD and solenoid designs with more reliable ESD system. Establish ESD valves and systems preventative maintenance plans. Develop automation procedure establishing accountability and documentation requirements.

CAUSAL FACTORS:

PEOPLE (ACTS): Use of Protective Methods: Inadequate use of safety systems

DATE: 14 Jan 2022

COUNTRY: USA

FUNCTION: Production

CAUSE: Exposure electrical

ACTIVITY: Maintenance, inspection, testing

RULE: Hot work

NARRATIVE:

An arc flash occurred while the IP was working on a recloser pole. A permanent impairment injury (PI) occurred.

WHAT WENT WRONG:

Arc flash.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

n/a

CAUSAL FACTORS:

PROCESS (CONDITIONS): Organizational: Inadequate work standards/procedures

NORTH AMERICA ONSHORE

DATE: 22 Apr 2022

COUNTRY: USA

FUNCTION: Production

CAUSE: Exposure noise, chemical, biological, vibration, extreme temperature

ACTIVITY: Production operations

RULE: Work authorization

NARRATIVE:

After performing clean out of transfer line, the IP was checking equipment. The IP was about to climb the ladder to the tank when his 4 gas monitor alarmed. The IP evacuated immediately and notified the supervisor.

WHAT WENT WRONG:

The IP was about to climb the ladder to the tank when his 4 gas monitor alarmed.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

- Lack of hazard recognition in Pre-Job planning and when performing task.
- Lack of communication from supervisor to crew (hazard awareness).
- No procedure available but should be.

CAUSAL FACTORS:

PEOPLE (ACTS): Following Procedures: Deviation unintentional (by individual or group)

PEOPLE (ACTS): Use of Protective Methods: Personal Protective Equipment not used or used improperly

PEOPLE (ACTS): Inattention/Lack of Awareness: Lack of attention/distracted by other concerns/stress

PROCESS (CONDITIONS): Protective Systems: Inadequate/defective warning systems/safety devices

PROCESS (CONDITIONS): Work Place Hazards: Hazardous atmosphere (explosive/toxic/asphyxiant)

PROCESS (CONDITIONS): Organizational: Inadequate hazard identification or risk assessment

PROCESS (CONDITIONS): Organizational: Inadequate communication

DATE: 22 Mar 2022

COUNTRY: USA

FUNCTION: Production

CAUSE: Exposure noise, chemical, biological, vibration, extreme temperature

ACTIVITY: Production operations

RULE: Unspecified

NARRATIVE:

Separator end cap failure.

WHAT WENT WRONG:

End cap on the separator had failed casing release.

Process Safety Fundamental: We walk the line.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

Develop suitable operational asset integrity program considering this failure.

NORTH AMERICA ONSHORE

CAUSAL FACTORS:

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate/defective tools/equipment/materials/products

PROCESS (CONDITIONS): Organizational: Inadequate work standards/procedures

DATE: 14 Jan 2022

COUNTRY: USA

FUNCTION: Production

CAUSE: Pressure release

ACTIVITY: Drilling, workover, well operations

RULE: Energy isolation

NARRATIVE:

Parked kill line hose.

WHAT WENT WRONG:

Kill line hose parted at a crimp connection.

Process Safety Fundamental: We apply procedures.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

Assure document pre-job inspection ensuring proper equipment is in place and to document reassigned wells/actions in a handover. Update procedures and ensure step to isolate pump tee and document.

CAUSAL FACTORS:

PEOPLE (ACTS): Use of Protective Methods: Equipment or materials not secured

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate/defective tools/equipment/materials/products

PROCESS (CONDITIONS): Organizational: Inadequate work standards/procedures

DATE: 04 Mar 2022

COUNTRY: USA

FUNCTION: Production

CAUSE: Pressure release

ACTIVITY: Drilling, workover, well operations

RULE: Unspecified

NARRATIVE:

Gas release.

WHAT WENT WRONG:

Natural gas release.

Process Safety Fundamental: We maintain safe isolation.

NORTH AMERICA ONSHORE

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

Develop freeze protect SOP with pressure limits and contingent solutions when pumping operations reach limits.
Update procedure.

CAUSAL FACTORS:

PROCESS (CONDITIONS): Organizational: Inadequate work standards/procedures

PROCESS (CONDITIONS): Organizational: Inadequate hazard identification or risk assessment

DATE: 06 Feb 2022

COUNTRY: USA

FUNCTION: Production

CAUSE: Struck by (not dropped object)

ACTIVITY: Drilling, workover, well operations

RULE: Line of fire

NARRATIVE:

Work String Failure

WHAT WENT WRONG:

Work string failure caused by drill pipe being separated below rotary and within BOP.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

Follow all pipe manufacturers procedures on make up and torque techniques and update operating practices to determine technique.

CAUSAL FACTORS:

PEOPLE (ACTS): Use of Protective Methods: Equipment or materials not secured

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate/defective tools/equipment/materials/products

PROCESS (CONDITIONS): Organizational: Inadequate work standards/procedures

DATE: 11 May 2022

COUNTRY: USA

FUNCTION: Production

CAUSE: Struck by (not dropped object)

ACTIVITY: Transport - Land

RULE: Driving

NARRATIVE:

Uncontrolled vehicle injury.

WHAT WENT WRONG:

Worker appears to have left his vehicle in gear after exiting.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

Reinforce the policy to engage the parking brake or turn the vehicle off prior exiting.

NORTH AMERICA ONSHORE

CAUSAL FACTORS:

PEOPLE (ACTS): Use of Tools, Equipment, Materials and Products: Improper use/position of tools/equipment/materials/products

PROCESS (CONDITIONS): Organizational: Inadequate work standards/procedures

DATE: 26 Sep 2022

COUNTRY: USA

FUNCTION: Production

CAUSE: Water related, drowning

ACTIVITY: Drilling, workover, well operations

RULE: Bypassing safety controls

NARRATIVE:

Personnel slipped into water in a frac pit.

WHAT WENT WRONG:

Worker slipped and slid down the pit wall into the water after collecting water samples.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

Develop standard procedure for frac pit water sampling. Incorporate best practices in future pit builds including rescue equipment.

CAUSAL FACTORS:

PROCESS (CONDITIONS): Organizational: Inadequate work standards/procedures

NORTH AMERICA OFFSHORE

DATE: 16 Aug 2022

COUNTRY: Mexico

FUNCTION: Drilling

CAUSE: Struck by (not dropped object)

ACTIVITY: Lifting, crane, rigging, deck operations

RULE: Safe mechanical lifting

NARRATIVE:

Elevator link described an oscillating movement when lifting top drive.

WHAT WENT WRONG:

- The steps in the operating procedure with casing running tool equipment are not described.
- In the manoeuvre the portable radius was not used to have good communication between the drilling cabin and the contractor manager.
- The task suspension policy is not effectively applied.
- Raised the Elevator Link with the Winch to remove the catwalk without realizing that the Elevator Link were together and uneven.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

- Update the procedure.
- Perform corrective maintenance and verify the proper functioning of the cabin communication system with the floor.
- Include in the JHA the identification of risks and barriers.
- Communication campaign on the stop the job policy.

CAUSAL FACTORS:

PEOPLE (ACTS): Following Procedures: Deviation unintentional (by individual or group)

PEOPLE (ACTS): Following Procedures: Improper lifting or loading

PEOPLE (ACTS): Use of Tools, Equipment, Materials and Products: Improper use/position of tools/equipment/materials/products

PEOPLE (ACTS): Use of Protective Methods: Failure to warn of hazard

PEOPLE (ACTS): Inattention/Lack of Awareness: Improper decision making or lack of judgment

PROCESS (CONDITIONS): Organizational: Inadequate work standards/procedures

PROCESS (CONDITIONS): Organizational: Inadequate hazard identification or risk assessment

NORTH AMERICA OFFSHORE

DATE: 17 Feb 2022

COUNTRY: USA

FUNCTION: Drilling

CAUSE: Dropped objects

ACTIVITY: Drilling, workover, well operations

RULE: Line of fire

NARRATIVE:

During a period of high winds, the wellsite sign became dislodged from the securing frame from which it was mounted on the antenna deck and fell to the deck below. There were no personnel in the area at the time; no other equipment was damaged. The 73lb. sign fell 15 feet, registering as high potential on the DROPS calculator.

WHAT WENT WRONG:

While preparing for an operation, the vessel was positioned to where high winds created optimal conditions to apply pressure to the sign. Additionally, securing hardware was present only around the edges of the sign, and its placement was in a location that provided better visibility and access but left it vulnerable to wind conditions.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

The combination of the frame design along with the hardware used was not adequate to ensure the sign was secured. The sign was constructed of thin sheet metal which is more vulnerable to wind effects and letters were secured by small wing nuts which present multiple hazards. The design also required two crew members to lift the sign up and over a handrail to change information.

CAUSAL FACTORS:

PROCESS (CONDITIONS): Protective Systems: Inadequate/defective guards or protective barriers

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate design/specification/management of change

PROCESS (CONDITIONS): Work Place Hazards: Storms or acts of nature

PROCESS (CONDITIONS): Organizational: Inadequate hazard identification or risk assessment

NORTH AMERICA OFFSHORE

DATE: 20 May 2022

COUNTRY: USA

FUNCTION: Drilling

CAUSE: Dropped objects

ACTIVITY: Drilling, workover, well operations

RULE: Line of fire

NARRATIVE:

While rigging up elevators to 16 foot bails in preparation of the 7 5/8 inch casing pull, the link tilt was kicked out away from well centre by inflating the link tilt air bag to install the secondary safety latch sensor on the elevators. The link tilt air bag ruptured resulting in the 16 foot bails coming to rest on the rig floor. A worker was tasked with installing the safety latch sensor and was positioned in between the elevators and the 7 5/8 inch casing stump. The casing elevators contacted the worker's right shin.

WHAT WENT WRONG:

When 16 foot bails are kicked out past the stops using the link tilt system, most equipment configurations will create a strain on the system that is beyond the rated capacity of the air bags.

The safety latch sensor was installed by the IP in a body position that created a line of fire risk.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

- The link tilt system was a pneumatic design which functioned by inflating the air bag. Other locations utilized a hydraulic link tilt with a greater operational capacity.
- The link tilt air bag was viewed as an expendable part. Previous failures were not recognized to have an HSE risk, and therefore, not investigated.
- Replacement parts were ordered through the rig contractor equipment acquisition process which does not interface with the Company inventory, inspection, and maintenance system. The operational life & parameters, and inspection & maintenance criteria for parts ordered this way are not tracked.
- An MOC was routed for an upgrade to a different casing elevator, however, the impact of this change was not fully understood – the new elevators require 16 foot bails which are too heavy for the link tilt air bag according to the load chart.
- To save time with the installation of the secondary safety latch sensor on the casing elevators, a deviation from the approved method of installation was made. The service provider trains personnel to install these sensors with the elevators placed on the ground, however, since the bails were already kicked out away from well-centre, the decision was made in the moment to complete the install with the system raised.

CAUSAL FACTORS:

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate design/specification/management of change

PROCESS (CONDITIONS): Organizational: Inadequate hazard identification or risk assessment

PROCESS (CONDITIONS): Organizational: Inadequate supervision

RUSSIA & CENTRAL ASIA ONSHORE

No high potential events reported.

RUSSIA & CENTRAL ASIA OFFSHORE

No high potential events reported.

SOUTH & CENTRAL AMERICA ONSHORE

DATE: 24 Feb 2022

COUNTRY: Argentina

FUNCTION: Drilling

CAUSE: Dropped objects

ACTIVITY: Drilling, workover, well operations

RULE: Bypassing safety controls

NARRATIVE:

When introducing the plug and perf (PnP) BHA to the Wireline lubricator for stage 25 of the well, the fishing neck got stuck inside the quick-lock device, and due to the over tension the mechanical weak point broke and the BHA fell to ground level (exposure of personnel).

WHAT WENT WRONG:

- Lack of skills and authorization: a new operator was appointed who did not have the skills for the task and the change was not reported.
- Lack of risk perception: Equipment operation at twice the speed established as safe. Operators are within the line of fire.
- Contradictory practice: The operating procedures have different values of speeds.
- Design problem of the connection with the quick pairing device.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

- Keep operating procedures updated and verify consistency between operator and contractor documents.
- Verify the designs of tools and quick connect devices in Rigless completion operations.
- Define a methodology for authorization of change of critical personnel in line with the required competencies.

CAUSAL FACTORS:

PEOPLE (ACTS): Following Procedures: Work or motion at improper speed

PEOPLE (ACTS): Use of Protective Methods: Failure to warn of hazard

PROCESS (CONDITIONS): Organizational: Inadequate training/competence

PROCESS (CONDITIONS): Organizational: Inadequate work standards/procedures

PROCESS (CONDITIONS): Organizational: Inadequate communication

SOUTH & CENTRAL AMERICA ONSHORE

DATE: 29 Sep 2022

COUNTRY: Argentina

FUNCTION: Drilling

CAUSE: Dropped objects

ACTIVITY: Drilling, workover, well operations

RULE: Line of fire

NARRATIVE:

Chain hoists were connected to the bridge crane, the basket was removed and the BOP skid was moved to the East side, causing the derailment and fall of the chain hoist.

WHAT WENT WRONG:

- Lack of work instructions.
- Lack of communication during simultaneous tasks.
- Lack of operational discipline.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

- Perform risk analysis of simultaneous tasks during assembly and disassembly.
- Guarantee fluid communication during simultaneous activities.
- Have specific instructions for the tasks that consider the risk analysis of simultaneous tasks.

CAUSAL FACTORS:

PROCESS (CONDITIONS): Organizational: Inadequate work standards/procedures

PROCESS (CONDITIONS): Organizational: Inadequate hazard identification or risk assessment

PROCESS (CONDITIONS): Organizational: Inadequate communication

PROCESS (CONDITIONS): Organizational: Poor leadership/organizational culture

DATE: 05 Mar 2022

COUNTRY: Argentina

FUNCTION: Production

CAUSE: Dropped objects

ACTIVITY: Drilling, workover, well operations

RULE: Bypassing safety controls

NARRATIVE:

During test activities with the PTS pump in a Pulling Rig, the person operating the machine detected that the lever brake system of the hoist did not fully respond and carried out 2 strokes using the equipment's socket/clutch to stop the downward movement.

This situation was observed by the Company Man who ordered to stop the task. In order to stop the task and leave the equipment in a safe condition, the operator decided to raise the pipe as much as possible to avoid leaving the pump resting on the bottom and place the pneumatic wedges.

As it rose, it reached the end of the stroke, causing the hoist to fall, reaching the bottom, resting on its side and bending the tubing.

SOUTH & CENTRAL AMERICA ONSHORE

WHAT WENT WRONG:

- Use of defective equipment: continuity of operation despite detecting failure in brakes.
- Lack of competence: machinist operator not qualified for the function.
- Inadequate work standard: it is not defined from what degree of lack of brake regulation the manoeuvre must be stopped. There is no instruction that defines the unsafe condition.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

- Place a physical indicator next to the brake lever that indicates the adjustment range at a glance.
- Develop a training plan for greater function in pulling and flushby teams.
- Reinforce training in the identification of critical security elements.
- Generate specific instructions on regulation and checking of the brake system.

CAUSAL FACTORS:

PEOPLE (ACTS): Following Procedures: Deviation unintentional (by individual or group)

PEOPLE (ACTS): Use of Protective Methods: Failure to warn of hazard

PEOPLE (ACTS): Use of Protective Methods: Inadequate use of safety systems

PROCESS (CONDITIONS): Protective Systems: Inadequate/defective warning systems/safety devices

PROCESS (CONDITIONS): Organizational: Inadequate training/competence

PROCESS (CONDITIONS): Organizational: Inadequate work standards/procedures

DATE: 10 Aug 2022

COUNTRY: Argentina

FUNCTION: Production

CAUSE: Explosion, fire or burns

ACTIVITY: Construction, commissioning, decommissioning

RULE: Hot work

NARRATIVE:

While performing tracing and thermal insulation activities over a tank located in a Crude Oil Plant, a sudden ignition and subsequent fire occurred. The fire was controlled by plant firefighting system and the Company's own firefighting team.

WHAT WENT WRONG:

- Inadequate Change Management (step from installation of works to operational).
- Vapour recovery unit was not operational.
- The Job Hazard Analysis was performed the day before the incident and did not include the current condition.
- Gas blanketing was not operational.
- The Area Authority did not participate in the planning meeting.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

- Adapt matrix to define team dedicated to commissioning and start-up.
- Comprehensive workshop between Operations, Facilities focusing on the pre-commissioning, commissioning and start-up stages.

SOUTH & CENTRAL AMERICA ONSHORE

CAUSAL FACTORS:

PEOPLE (ACTS): Following Procedures: Deviation unintentional (by individual or group)

PEOPLE (ACTS): Use of Tools, Equipment, Materials and Products: Improper use/position of tools/equipment/materials/products

PEOPLE (ACTS): Use of Protective Methods: Inadequate use of safety systems

PEOPLE (ACTS): Use of Protective Methods: Equipment or materials not secured

PEOPLE (ACTS): Inattention/Lack of Awareness: Improper decision making or lack of judgment

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate design/specification/management of change

PROCESS (CONDITIONS): Work Place Hazards: Hazardous atmosphere (explosive/toxic/asphyxiant)

PROCESS (CONDITIONS): Organizational: Inadequate hazard identification or risk assessment

PROCESS (CONDITIONS): Organizational: Inadequate supervision

PROCESS (CONDITIONS): Organizational: Poor leadership/organizational culture

PROCESS (CONDITIONS): Organizational: Failure to report/learn from events

DATE: 07 Apr 2022

COUNTRY: Argentina

FUNCTION: Production

CAUSE: Exposure electrical

ACTIVITY: Maintenance, inspection, testing

RULE: Energy isolation

NARRATIVE:

During well telemetry maintenance tasks, a short circuit occurred in the frequency inverter in the electrical power input terminal block.

WHAT WENT WRONG:

Job control process was not applied: The task was not included in the schedule. It was not correctly categorized, therefore a work permit was not processed.

A risk analysis was not performed for the task and there was no procedure to perform it safely.

Inadequate barrier/protection: lack of protection elements to avoid accidental contacts or electrical arc.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

- Guarantee that all personnel and contractors know and apply the control of work process.
- Verify the inclusion of all tasks in the programming of the asset/facility.
- Verify protections on panels to prevent contacts or electric arc.

CAUSAL FACTORS:

PEOPLE (ACTS): Following Procedures: Deviation unintentional (by individual or group)

PROCESS (CONDITIONS): Protective Systems: Inadequate/defective guards or protective barriers

PROCESS (CONDITIONS): Organizational: Inadequate work standards/procedures

PROCESS (CONDITIONS): Organizational: Inadequate communication

SOUTH & CENTRAL AMERICA ONSHORE

DATE: 18 May 2022

COUNTRY: Brazil

FUNCTION: Drilling

CAUSE: Dropped objects

ACTIVITY: Maintenance, inspection, testing

RULE: Other

NARRATIVE:

During the monthly/weekly inspection of the lifeboats and the rescue boat (scope of the preventive maintenance program), in the test of the descent brake of the rescue boat, the davit winch cable broke and the boat fell into the sea from an approximate height of 20 metres. The test consists of releasing the boat with braking after approximately 5 metres of descent by gravity. When braking, the cable broke. The boat was unmanned, as per the test procedure. There was no risk of hitting people. There was no rupture of fuel or lubricant tanks.

WHAT WENT WRONG:

- Test execution with descent speed higher than specified in the manufacturer's manual and IMO.
- Dynamic brake failure to maintain the design speed.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

- Evaluate the inclusion of a task to measure the descent speed of the rescue boat with a frequency of less than 5 years.
- Apply the compliance with the Procedure Verification tool to verify that the procedures for operating safeguard equipment are adequate and that they are being complied with.

CAUSAL FACTORS:

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate design/specification/management of change

DATE: 18 Feb 2022

COUNTRY: Brazil

FUNCTION: Production

CAUSE: Pressure release

ACTIVITY: Maintenance, inspection, testing

RULE: Bypassing safety controls

NARRATIVE:

After replacing the oxygen cylinder of the oxyacetylene set, when the worker opened the cylinder valve, there was an abrupt increase in pressure. At that moment, the pressure regulator and flame arrester could not withstand this sudden increase, causing a burst and spring projection against the workshop wall. The worker was not hit, due to his position on the opposite side of the projection.

WHAT WENT WRONG:

- Maintenance failure: cylinders reconfirmed without internal cleaning control and procedures.
- Maintenance failure: no maintenance plan on the pressure regulator.
- Procedure failure: the procedure did not provide for a welding service with an oxyacetylene set.
- Human Factors (behavioural): lack of attention, stress, worry, etc.

SOUTH & CENTRAL AMERICA ONSHORE

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

- Request the periodic inspection certificate and the product analysis certificate from the oxygen cylinder supplier.
- Create a maintenance plan for pressure regulators.
- Review the contracted company's procedure, including the activities and risk analysis of the oxyacetylene welding process.
- Carry out training with the entire team involved and demonstrate it with an attendance list.

CAUSAL FACTORS:

PEOPLE (ACTS): Use of Tools, Equipment, Materials and Products: Servicing of energized equipment/inadequate energy isolation

PEOPLE (ACTS): Use of Protective Methods: Failure to warn of hazard

PEOPLE (ACTS): Inattention/Lack of Awareness: Lack of attention/distracted by other concerns/stress

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate maintenance/inspection/testing

PROCESS (CONDITIONS): Organizational: Inadequate hazard identification or risk assessment

DATE: 04 Sep 2022

COUNTRY: Peru

FUNCTION: Production

CAUSE: Explosion, fire or burns

ACTIVITY: Production operations

RULE: Work authorization

NARRATIVE:

During maintenance work on a 3" automatic valve, located at the NGL outlet of a condensate stabilizer (on standby) there was a leak of condensed liquid. During the event, there was no damage to people or assets. This event was classified as a Tier 2 PSE.

WHAT WENT WRONG:

PERSONAL FACTORS

- Lack of risk awareness. There was no evidence of the task's planning/scope process performed by either the planner or the operator. During planning, it is not detected that the valve to be intervened does not have blocking valves.
- Inadequate discipline. Work Permit with an isolation certificate that does not correspond. The lack of adequate isolation certificate is not detected neither by operator nor by local area authority.

LABOR / SYSTEM FACTORS

- Inadequate work scheduling/planning. The planning/scope process was performed by either the planner or the operator.
- Normalization of deviation from a procedure. Upon the change in the scope of the task, the local area authority nor the operator required to suspend the task and renew the ATS.
- Inadequate identification of risks in the design. Valve to intervene does not have blocking valves.
- Inadequate assessment of preventive maintenance needs. The preventive maintenance program is not executed.

Process Safety Fundamentals: We maintain safe isolation. We respect hazards.

SOUTH & CENTRAL AMERICA ONSHORE

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

- Carry out reinforcement training in compliance with the Life Saving Rules.
- Ensure that Contractors maintenance personnel have the appropriate competence for the work to be carried out in the plant.
- Carry out training campaigns on hazards and risk of the plant process and Stop the Job Policy for maintenance personnel.
- Design control valve maintenance strategy.

CAUSAL FACTORS:

PEOPLE (ACTS): Following Procedures: Deviation unintentional (by individual or group)

PEOPLE (ACTS): Inattention/Lack of Awareness: Improper decision making or lack of judgment

PROCESS (CONDITIONS): Organizational: Inadequate training/competence

PROCESS (CONDITIONS): Organizational: Inadequate supervision

DATE: 08 May 2022

COUNTRY: Peru

FUNCTION: Production

CAUSE: Struck by (not dropped object)

ACTIVITY: Production operations

RULE: Bypassing safety controls

NARRATIVE:

During river water filling operation of tank, there was a loss of containment in the walls of the riveted tank, causing the tank to collapse.

WHAT WENT WRONG:

PERSONAL FACTORS

- Inadequate discipline. Normalization of the tank overfill deviation.

LABOR FACTORS

- Inadequate Management of Change. Change of service of the tank / No evidence of MoC for the dosing of chemicals to the water. The design of the tank was not verified when it was incorporated into the operation. Tank does not meet the applicable manufacturing standard for this service. The risk of changing the density of the water due to the accumulation of solids inside the tank was not identified.
- Inadequate assessment of preventive maintenance needs. There is no preventive maintenance program and no critical evaluation registered in company system for the asset. Equipment was not available for inspection.
- Lack of supervision. Normalization of deviation. The tank's condition as rejected was not kept after the Integrity inspection; the defective equipment was operated; tank repair was not prioritized.
- Inadequate assessment of needs and risks. The tank is not registered as a safety critical element and does not have a criticality analysis in the company system.
- Inadequate work procedure. There was no specific procedure or limits defined for the filling operation of the tank.

SOUTH & CENTRAL AMERICA ONSHORE

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

- Ensure the correct execution of preventive maintenance programs.
- Ensure the availability of equipment when requested for scheduled maintenance.
- Report instrument control and safety systems failures.
- Ensure management of change when tanks are assigned for a new use other than the one for which it was designed.

CAUSAL FACTORS:

PEOPLE (ACTS): Use of Tools, Equipment, Materials and Products: Improper use/position of tools/equipment/materials/products

PROCESS (CONDITIONS): Protective Systems: Inadequate/defective warning systems/safety devices

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate design/specification/management of change

PROCESS (CONDITIONS): Organizational: Inadequate hazard identification or risk assessment

DATE: 30 May 2022

COUNTRY: Peru

FUNCTION: Production

CAUSE: Unspecified - Other

ACTIVITY: Production operations

RULE: Bypassing safety controls

NARRATIVE:

During normal operation at the Gas Plant, there was a plant shutdown due to the activation of the safety system.

WHAT WENT WRONG:

PERSONAL FACTORS

- Inadequate discipline. The communication sequence that must be followed according to the procedure for enabling and disabling a safety critical element was not followed.

LABOR / SYSTEM FACTORS

- Inadequate communication. Knowledge of the inhibition procedure at Operators and Contractors level is not ensured.
- Inadequate identification of the failure mode. In the safety shutdown system design, the inherent risk of connecting different high-level signals and others (on which frequent maintenance is performed) to shared or common communication cards was not identified.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

- Ensure the correct execution of the safety critical elements inhibition procedures.
- Correctly assess the risks inherent to an activity or task during the execution of preventive maintenance.
- Adequately manage changes in projects for the expansion or modification of existing facilities.

SOUTH & CENTRAL AMERICA ONSHORE

CAUSAL FACTORS:

PEOPLE (ACTS): Following Procedures: Deviation unintentional (by individual or group)

PEOPLE (ACTS): Use of Protective Methods: Failure to warn of hazard

PROCESS (CONDITIONS): Protective Systems: Inadequate/defective warning systems/safety devices

PROCESS (CONDITIONS): Organizational: Inadequate hazard identification or risk assessment

PROCESS (CONDITIONS): Organizational: Inadequate communication

DATE: 18 Jan 2022

COUNTRY: Trinidad & Tobago

FUNCTION: Construction

CAUSE: Exposure electrical

ACTIVITY: Lifting, crane, rigging, deck operations

RULE: Bypassing safety controls

NARRATIVE:

Worker felt an electrical shock while attempting to relocate a scaffold rack stored on the platform deck. Investigation identified that the scaffolding was rested on two 480V electrical cables that ran the perimeter of the deck. The insulation on one cable was broken, exposing the electrical conductor.

WHAT WENT WRONG:

Temporary cable was laid across the deck at the contractor's fabrication yard prior to sail-away. The improper installation was identified during mechanical walkdowns; however, the handwritten punch list notes were not clearly documented therefore not completed.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

- Ensure lay down areas are clear when conducting lifting activities and that electrical cables are appropriately protected, identified, and labelled.
- Ensure there is an active and effective process for documenting punch List actions.

CAUSAL FACTORS:

PROCESS (CONDITIONS): Protective Systems: Inadequate/defective guards or protective barriers

SOUTH & CENTRAL AMERICA OFFSHORE

DATE: 01 May 2022

COUNTRY: Brazil

FUNCTION: Production

CAUSE: Dropped objects

ACTIVITY: Lifting, crane, rigging, deck operations

RULE: Safe mechanical lifting

NARRATIVE:

After completion of activities under the scope of a preventive maintenance on Deck Crane B, an employee was preparing to lower materials from the top of the crane. A rope was tied with a single knot to the handles of a tool bag and a plastic bucket, mainly containing tools. At the beginning of the descent movement, when the set was at the crane's upper platform level, it bumped at its guardrail and the knot came loose and the materials fell to the Main Deck, which had not been barriered-off for this crane maintenance. The location of the Main Deck where the objects hit was close to an escape route, being an area of regular and frequent passing of people. Height fall was measured as 17.42 meters and load was weighted as 9.38 kg, resulting in a fall kinetic energy of 1,603 Joules.

WHAT WENT WRONG:

- The requirements for barriers and securing tools were not followed.
- The employees did not foresee the possibility of the knot being untied. This had been done many times before and had been a standard way of transporting tools and equipment when working at the crane.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

- Create standard for transporting tools between floor levels.
- Reinforce the improvement of isolation barriers around possible area of impact of falling objects in all teams.
- Remove all ropes stored in the crane that are used to lift manual tools/equipment. All manual tools or equipment to be used in any maintenance work at the crane to be transferred in adequate tool bag.
- Reinforce the use of 'Drops prevention' tools to perform any approved work when there is a risk of falling objects above 2 metres and properly secure tools and equipment when there is a need to transfer them up/down the unit levels. Confirm that control of "drops prevention" tools are in place, i.e.: logbook, to minimise the risk of potential dropped object.

CAUSAL FACTORS:

PEOPLE (ACTS): Following Procedures: Deviation unintentional (by individual or group)

PEOPLE (ACTS): Use of Tools, Equipment, Materials and Products: Improper use/position of tools/equipment/materials/products

PROCESS (CONDITIONS): Organizational: Inadequate hazard identification or risk assessment

PROCESS (CONDITIONS): Organizational: Inadequate communication

PROCESS (CONDITIONS): Organizational: Poor leadership/organizational culture

SOUTH & CENTRAL AMERICA OFFSHORE

DATE: 25 Mar 2022

COUNTRY: Brazil

FUNCTION: Production

CAUSE: Dropped objects

ACTIVITY: Maintenance, inspection, testing

RULE: Line of fire

NARRATIVE:

Scaffolding tube (4m long) fell from a height of 15 metres with a weight of 14.4 kg onto the main deck in the bow below the flare superstructure. The tube first collided with a guardrail that was on the floor (waiting for installation), hit the floor and, when lying on the deck, brushed against a worker who was in the area. No injury was found to the worker.

WHAT WENT WRONG:

- Inadequate risk identification, assessment, consideration, and mitigation.
- Failure to guarantee the performance of activities in accordance with the established requirements.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

- Install restraints at the ends of drawers that have the potential to fall from the level or lock the scaffolding tubes in the drawer to control the risk of falling.
- Insert guidance in the contractor's operating procedure regarding safety requirements regarding the storage of scaffolding materials, with emphasis on the assembly of scaffolding drawers. Containment must be installed in the drawer in places with the potential for a fall or to lock the tubes to prevent movement and drops.
- Conduct training with the contractor's executing team involved in the procedure mentioned in action 2.
- Evaluate the inclusion of guidance in the corporate procedure (Work at height - scaffolding) on HSE requirements regarding the storage of scaffolding materials. Containment must be installed in the drawer in places with the potential for a fall or to lock the tubes to prevent movement and drops.
- Carry out an inspection on the decks of the platform turret, to collect materials left loose/at risk of falling.
- Reinforce guidance with the contractor's Work authorization applicants regarding the requirement of the Safety Manual. At the end of the work, the requester must ensure that the place where the service was performed is in perfect conditions of order, cleanliness and tidiness and that the waste generated was properly disposed of.

CAUSAL FACTORS:

PEOPLE (ACTS): Following Procedures: Deviation unintentional (by individual or group)

PEOPLE (ACTS): Use of Tools, Equipment, Materials and Products: Improper use/position of tools/equipment/materials/products

PEOPLE (ACTS): Use of Protective Methods: Failure to warn of hazard

PEOPLE (ACTS): Inattention/Lack of Awareness: Improper decision making or lack of judgment

PROCESS (CONDITIONS): Protective Systems: Inadequate/defective guards or protective barriers

SOUTH & CENTRAL AMERICA OFFSHORE

DATE: 10 Aug 2022

COUNTRY: Brazil

FUNCTION: Production

CAUSE: Falls from height

ACTIVITY: Transport - Water, incl. marine activity

RULE: Other

NARRATIVE:

An offshore support vessel was near a drilling rig in adverse sea conditions. The vessel's captain was positioned in the forward command, and moved towards the stairs, which gave access to the lower floor. When the vessel rocked he lost his balance going towards the stairs, falling backwards and sliding to the end of the stairs (height 3.2 metres) on the lower floor.

WHAT WENT WRONG:

- Absence of auxiliary safety device for hand support.
- Lack of risk perception focused on safe positioning.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

- Carry out a study on the entire fleet to install handholds on the consoles of vessels that do not have such a device.
- Carry out training based on the study of safe positioning in the houses/Walkway with the entire fleet.

CAUSAL FACTORS:

PEOPLE (ACTS): Inattention/Lack of Awareness: Lack of attention/distracted by other concerns/stress

PROCESS (CONDITIONS): Protective Systems: Inadequate/defective guards or protective barriers

PROCESS (CONDITIONS): Protective Systems: Inadequate security provisions or systems

PROCESS (CONDITIONS): Organizational: Inadequate work standards/procedures

PROCESS (CONDITIONS): Organizational: Inadequate supervision

SOUTH & CENTRAL AMERICA OFFSHORE

DATE: 08 Mar 2022

COUNTRY: Guyana

FUNCTION: Drilling

CAUSE: Struck by (not dropped object)

ACTIVITY: Maintenance, inspection, testing

RULE: Line of fire

NARRATIVE:

Individual sustained hand injuries when struck by piston during BOP Maintenance Operations.

WHAT WENT WRONG:

Failure to identify stored energy hazard. Inadequate identification of critical task step. Work plan inadequate and lacking task specific work plan. Inadequate verification of proper use and reinforcement of compliance with procedures by management.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

Reinforce hazard identification and adherence to procedures.

CAUSAL FACTORS:

PEOPLE (ACTS): Following Procedures: Improper position (in the line of fire)

PEOPLE (ACTS): Use of Tools, Equipment, Materials and Products: Servicing of energized equipment/inadequate energy isolation

PROCESS (CONDITIONS): Organizational: Inadequate work standards/procedures

PROCESS (CONDITIONS): Organizational: Inadequate hazard identification or risk assessment

PROCESS (CONDITIONS): Organizational: Inadequate supervision

PROCESS (CONDITIONS): Organizational: Failure to report/learn from events

For the full analysis of 2022 results,
as well as all fatal incident reports
submitted to IOGP since 1991,
visit IOGP's data website at:
<https://data.iogp.org>

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